Engineering Germany's Industrial Success:

Professional Societies and the Social Dynamics of Innovation*

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Abstract

Germany was among the few countries that reached the global technological frontier, rising from an industrial laggard in 1850 to a world leader in chemistry and electrical machinery by 1913. What role did the organization of its engineers play in this catch-up? I study the German Engineering Society, whose local chapters held regular meetings at which engineers and owners of different firms discussed technical papers and visited plants. Exploiting the staggered founding of chapters between 1880 and 1912 in a difference-in-differences design, I find that the annual number of high-value patents in a chapter's region rose by 4.8 percent after it opened. Pre-trends are flat, and the effect is robust to controlling for prior industrialization. The increase was concentrated in industrial fields in which Society members worked. After an opening, high-value patents became more similar to earlier patents from their own district, consistent with knowledge exchange among local engineers. A century later, counties near the chapter towns still patent more.

JEL codes: N13, N93, O31, O33.
Keywords: engineers; associations; patents; Imperial Germany.

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1. Introduction

Germany was among the few countries that reached the technological frontier, and by 1913 it had overtaken Britain in manufacturing output (Bairoch, 1982). The literature attributes this catch-up to education (Becker et al., 2011; Cinnirella and Streb, 2017; Dittmar and Meisenzahl, 2022) and to the patent law of 1877 (Donges and Streb, 2023). Engineers were central to both: the technical schools trained them, and the patent law protected their inventions. Yet much of the knowledge that engineers used was held in firms and passed on from peer to peer (Vincenti, 1990), and such exchange is more likely among nearby inventors (Jaffe et al., 1993). This paper studies a professional society that brought together the engineers of different firms and shows that it raised invention in the places it reached.

The German Engineering Society (Verein Deutscher Ingenieure, hereafter the Society) organized such exchange through regional chapters. Each chapter covered a territory and held regular meetings in its host town, where engineers and owners of different firms discussed technical papers and visited plants. I use the staggered opening of 22 chapters between 1880 and 1912 to estimate the effect of a chapter on invention in its territory. I measure invention by high-value patents, the patents that their holders renewed for at least ten years. After a chapter opened, places in its territory obtained 4.8 percent more high-value patents per year.

The Engineering Society was founded in 1856 by young engineers, who wanted to "encourage and teach each other" (Gispen, 1989). It admitted owners and salaried engineers alike, never required a diploma, and grew from 309 members in 1857 to about 27,000 in 1925; its directory of 1900 lists Robert Bosch next to ordinary works engineers. Its members met in self-governing local chapters, which Theodor Peters, who ran the Society's office, described as places "where the employee is perfectly free to engage his boss on purely technical matters" (Gispen, 1989). Members founded new chapters where the existing ones were far away or had grown large: 25 chapters existed by 1877, when the patent data begin, and the 22 chapters of the design opened between 1880 and 1912, from East Prussia and Schleswig-

Holstein to the Mosel. A single patent law applied throughout Germany, and the design exploits differences in whether, and since when, the engineers of a place had a chapter in which to meet.

To measure invention, I use the 25,800 high-value patents in the data of Streb et al. (2006): patents granted between 1877 and 1918 to holders in Germany and renewed for at least ten years. Because renewal fees rose each year, holders kept only valuable inventions. To place the Society's members, I digitize its membership directories of 1900 and 1925, which list each member with town, position, employer, and chapter (15,513 entries in 1900 and 27,176 in 1925), together with the size of each chapter since 1857. The occupational censuses of 1882, 1895, and 1907 measure the local industrial structure; for Prussia, I use the ifo Prussian Economic History Database (Becker et al., 2014), and for the other territories, I digitize the census volumes.

I estimate the effect with a staggered difference-in-differences design across the 493 places in the territories of the 22 new chapters, comparing places whose chapter had opened with places whose chapter had not yet opened (Callaway and Sant'Anna, 2021) and controlling for railway kilometers, the population of 1871, and the population density of 1900.[1] The outcome is the log number of high-value patents a place obtained in a year, from 1877 to 1918.

After their chapter opened, places obtained 4.8 percent more high-value patents per year ($p < 0.01$). Over the 16 years of the event-study window (years 0 to 15 after an opening; Figure 5), the effect averages 3.0 percent per year; summed over these years, the additional patents equal about half of a place's annual high-value patents.

The main threat to identification is that chapters opened where invention was about to rise for other reasons. Before an opening, places in the territory of a new chapter and not-yet-treated places followed parallel trends: the pre-founding coefficients average 0.6 percent and are jointly insignificant. The level of patenting and the distance to the nearest chapter

[1] A place is a statistical unit of 1900, a district or its equivalent outside Prussia.

predicted where chapters opened, but recent growth in patenting did not. Controlling for the industrial structure of 1882 yields estimates between 3.0 and 5.5 percent. For the chapters that opened after the census of 1895 and the directory of 1900, controlling for the industry of 1895 or for the stock of Society members yields estimates that are as large or larger. Because every place eventually received a chapter, the comparison group shrinks over time. Designs that hold it fixed cover mainly the first decade after an opening and put the effect over those years at about 1 percent, statistically indistinguishable from the 2.3 percent of the main design over the same years.

The event study shows that the effect built up gradually. The coefficients lie between -2.1 and 1.8 percent before founding and rise to between 4.6 and 9.3 percent in years 7 to 15 after it. This delay is consistent with the years needed to develop a design and with the year between the application for a patent and its grant. It is also what one would expect if the chapters worked through relations among engineers that took time to form.

The increase in high-value patents was concentrated in the fields in which the Society's members worked. Of the 89 patent classes, I classify 50 as industrial (the fields of the Second Industrial Revolution, such as machinery and electrical engineering) and 38 as traditional (mostly crafts and consumer goods). In the industrial classes, high-value patents rose by 3.4 percent per year after an opening; in the traditional classes, the estimate is 1.2 percent and insignificant. The effect was also concentrated in places that were already industrial in 1882, reaching 10.5 percent per year in the most industrial third. These results suggest that the chapters raised invention where industry and engineers were already present.

There is no evidence that the chapters attracted engineers or industry. Where a chapter opened between 1900 and 1925, Society members per inhabitant, employment in the engineering industries, and the industry share did not grow significantly faster than elsewhere. Engineering capacity without a chapter did not raise invention either: places 10 to 50 km from one of the 24 state machine-building schools that opened during the period, which trained technicians but did not bring a chapter, did not obtain significantly more high-value

patents than places 50 to 100 km away. The increase therefore appears to have come from engineers who were already in place.

If chapters spread knowledge among local engineers, new patents in chapter places should draw more on earlier local work. To test this prediction, I use the text embeddings that Google Patents publishes for almost every German patent of the period (Abi Younes and de Rassenfosse, 2024) and measure similarity as in studies of US patents (Kelly et al., 2021; Arts et al., 2021). After an opening, high-value patents became about 3 percent more similar to the patents of their own place in the five preceding years, relative to patents of the same class in other territories. Their similarity to foreign patents and to patents of other territories, which serve as placebo outcomes, did not change. This indicates that, after an opening, inventors in chapter places drew more on earlier work from their own place, consistent with knowledge exchange among local engineers.

Invention remained concentrated around the chapter towns long after 1918. Using patent data from PatentCity (Bergeaud and Verluise, 2024), I compare West German counties in an inner ring 10 to 50 km from each chapter town with counties in an outer ring 50 to 100 km away, controlling for each county's population in 1871. Since the 1920s, counties in the inner ring have held 24 to 42 log points more patents than those in the outer ring. Around cities without a chapter, the same comparison shows fewer patents near the city. These comparisons are descriptive, but they are consistent with a lasting advantage of the regions that organized their engineers early.

Taken together, the results suggest that the chapters raised invention by bringing together engineers who were already in place, and that the effect was largest where industry was dense and in the fields in which the members worked.

This paper makes three contributions. First, it separates how engineers were organized from how many there were. A large literature shows that engineers were central to modern growth: Maloney and Valencia Caicedo (2022) show that the density of engineers during the Second Industrial Revolution is related to income across US counties and the countries

of the Americas today, and Hanlon (2025) shows that the rise of the engineering profession changed how invention was done during the Industrial Revolution.[2] Because the number of engineers and their organization often changed together, the two are hard to separate. The dated chapter openings, together with the membership directories and the censuses, make it possible to do so.

Second, it adds a local institution to the inputs that studies of industrialization link to invention and growth, such as upper-tail human capital in France (Squicciarini and Voigtländer, 2015) and, in Germany, schooling, research universities, and legal reform, most of them provided by the state (Becker et al., 2011; Cinnirella and Streb, 2017; Becker et al., 2023; Dittmar and Meisenzahl, 2022; Donges et al., 2023). Unlike these inputs, a chapter was financed by its members and reached only the engineers who could attend its meetings.

Third, it relates to studies of associations and the spread of technical knowledge. Historical studies argue that learned and economic societies spread useful knowledge (Mokyr, 2002), and studies of social capital relate civic associations to better regional government (Putnam et al., 1993), trust to more regional innovation (Akçomak and ter Weel, 2009), and dense associational networks to the faster rise of the Nazi Party (Satyanath et al., 2017), mostly in cross-sections. By comparing dated openings with not-yet-treated places, this paper removes fixed differences between places with dense and sparse associations. Closest in design, Andrews (2026) shows that invention fell where state prohibition laws closed the saloons, places of informal exchange; this paper studies the reverse change, the opening of a formal meeting place for engineers. Closest in setting, Cinnirella et al. (2025) show that the economic societies of the late eighteenth century lowered the cost of access to useful knowledge: German regions with more society members were more innovative in the late nineteenth century, and regions with members of the same society patented more similarly. Both papers point to the same channel, the local exchange of knowledge among members.[3]

[2]Kelly et al. (2023) relate British industrialization to the mechanical skills of artisans, and Rubens (2023) shows that the first graduates of American mining schools raised the productivity of the coal mines they managed.

[3]Places with more members of the economic societies also had more members of the Engineering Society

2. Historical Background

2.1. Technical Education and Engineers

Technical education in Germany grew out of the reforms that Prussia adopted after its defeat by Napoleon: it freed trade in 1810, rebuilt secondary and higher education around the classical Gymnasium and the new University of Berlin, and expanded its elementary schools (Becker et al., 2011; Becker et al., 2023). Because these reforms kept practical knowledge out of general education, the trade department built technical schools of its own from 1820.

The states built these schools before there was an industry to employ their graduates or a profession to organize them. In Prussia, Peter Beuth founded the Berlin Industrial Institute (Gewerbeinstitut) in 1821 as "a humble school for builders, chemists, and machine builders," fed by a network of provincial trade schools (Gispen, 1989). Across the German states, more than a dozen polytechnic schools were founded between 1803 and 1830, partly on the model of the French École polytechnique, among them those of Karlsruhe, Munich, and Stuttgart (Homburg, 1997).

Over the century, these schools rose to academic rank, and in 1899 the technical universities (technische Hochschulen) won the right to award doctorates. The title of engineer (Ingenieur), however, required no qualification and was not protected by law until 1965. The technical universities counted close to 15,000 students in 1900, and around 1914 contemporary estimates put the number of engineers at 100,000 to 150,000 (Gispen, 1989).

Most engineers worked in firms, and in machine building, the Society's core industry, design and invention were closely linked. The industry relied on "a constant flow of minor, unpatented improvements and innovations." In a dispute of 1909 over the inventions of employed engineers, an employer held that "designing and inventing are identical," and the machine builders' association estimated that up to a quarter of all German patents were for mechanical designs (Gispen, 1989).

in 1900 (Appendix F.3), which I read only as indicative evidence that the two networks overlapped in space.

The patent law of 1877 gave Imperial Germany a single patent system, which was revised in 1891. Applications rose from about 4,000 a year in the late 1870s to 40,000–50,000 before the First World War. The patent went to the first applicant, and employment contracts often assigned the inventions of employees to the employer, so German patents of the period name the holder, often a firm (Gispen, 1989). Two features of this history matter for the design: the states set the supply of engineers, and one patent law applied everywhere. The design uses variation in whether, and since when, the engineers of a region had a chapter in which to meet.

2.2. The Society and Its Chapters

The Engineering Society was founded in May 1856 by 23 men, most of them former members of a student club in Berlin, together with one of their former teachers (Peters, 1912). They wanted an association "that allows them to encourage and teach each other as well as to combine their forces for the development of national industry, even though they were living in the farthest corners of the fatherland," and the Society was national by design and "scrupulously 'unpolitical' " (Gispen, 1989). In the words of Hortleder (1970), it was founded "as an association for the advancement of technology, not as an interest organization for the benefit of the engineers."

The Society was open to all who worked in technology. Ordinary membership was "unconditionally open to: (a) practicing engineers, (b) teachers of technology and technical sciences, (c) owners and managers of technical establishments, (d) non-engineers," and a diploma was never a condition of admission. It grew from 309 members in 1857 to 2,981 in 1876 and, by its own count, to about 12,000 in 1897. In 1876 owners made up 26 percent of the members, directors 16 percent, and salaried designers, superintendents, and lower managers 25 percent (Gispen, 1989).

The chapters (Bezirksvereine) were the Society's regional units. They governed themselves, with unpaid chairmen and committees that were "usually notable entrepreneurs,

managers, or respected employees," and their meetings "served both as social events and as occasions for professional exchange, such as the discussion of shop experiences, the presentation of papers, or visits to technical installations." Theodor Peters, who ran the Society's office, described the chapters as places "where the employee is perfectly free to engage his boss on purely technical matters" (Gispen, 1989).

A chapter added regular personal contact among the engineers of different firms in a region. Because the Society's journal, which printed the proceedings of the chapters, reached every member from 1857 on wherever he lived, a local effect of an opening is likely to work through the meetings.

2.3. The Spread of Chapters

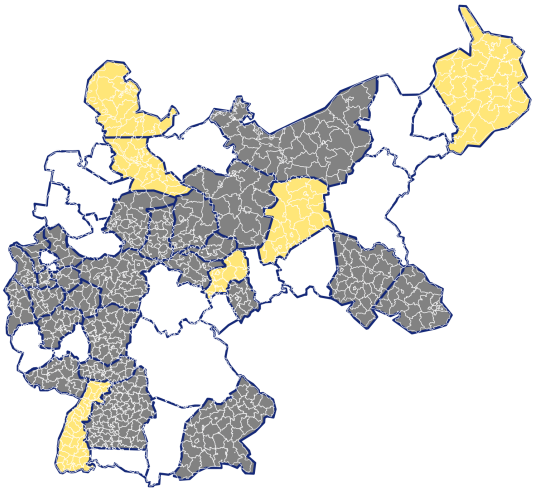
Fifty chapters, not counting that of Alsace-Lorraine, were founded between 1856 and 1924: 25 by 1877, when the patent law took effect; 22 between 1880 and 1912, in 16 founding years; and three after 1918.[4] The chapters founded by 1877 covered most of the early industrial regions of Germany. The 22 chapters of 1880–1912 opened all across Germany, from East Prussia and Schleswig-Holstein in 1880 to the Bodensee in 1908 and the Mosel in 1912 (Figure 1).

Members founded new chapters where the existing ones were far away or had grown large. The Chemnitz chapter, refounded in 1886, split off from the Saxon chapter in Leipzig, which had covered all of Saxony since 1882, and the Dresden chapter of 1897 split off from the Chemnitz chapter (Peters, 1912). Chapters founded later started with more members, although they opened in smaller towns. The median founding roll was 23 members before 1880, 87 in 1880–99, and 128 after 1900 (Appendix D.2).

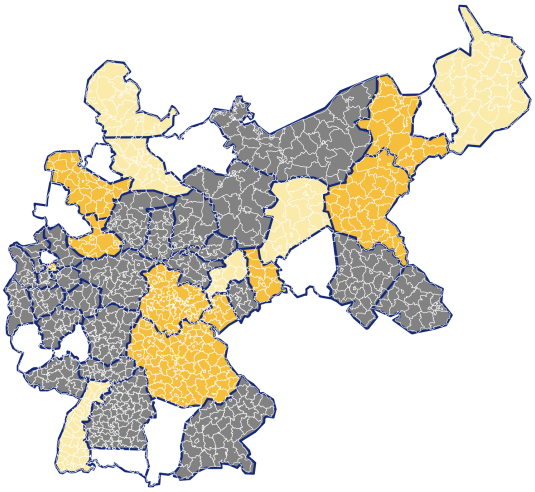
Members who lived far from a chapter town, the seat of a chapter, belonged to a chapter less often. In the directory of 1900, 10 percent of the members within 5 km of a chapter

[4]The three chapters founded after 1918 are Elbing and Lübeck (1921) and Osnabrück (1924). Some chapters lapsed and were refounded, so counts by founding date differ from counts of active chapters (Appendix A.1). The chapter of Alsace-Lorraine, founded in 1895, whose territory was ceded in 1919, is excluded throughout.

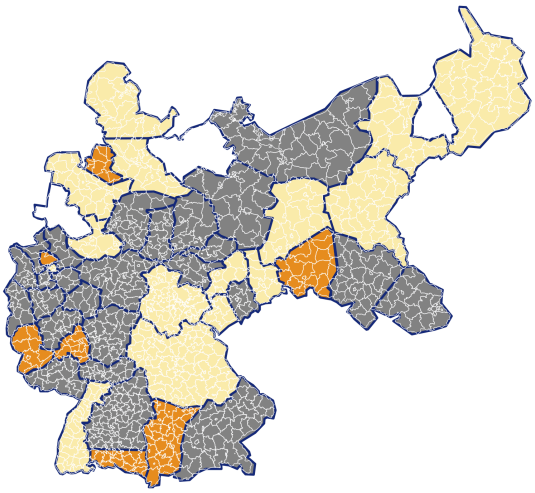
Figure 1. Chapter Territories by Founding Period



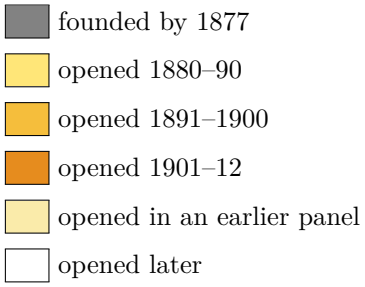
(a) By 1890



(b) By 1900



(c) By 1918



Notes: Chapter territories at their 1925 shape on the statistical units of 1900, shaded by the period in which the chapter opened; panels (b) and (c) show the chapters opened in the earlier panels in a pale shade. White: territories whose chapter opened after the year of the panel; in panel (c), those of the chapters of Elbing and Lübeck (1921) and Osnabrück (1924), which opened after the patent data end in 1918. Lines: territory borders. Territories and founding dates from the Society's yearbooks (Appendix A.1).

town belonged to no chapter, against 24 percent of those 30 to 50 km away and 31 percent of those more than 75 km away. Dues were 15 marks a year after 1873, about 3 percent of an industrial worker's annual pay in 1871 and 1.2 to 1.4 percent in 1913.[5] Until 1911, members could choose whether to join a chapter.

An opening brought the members of a territory closer to a chapter town. The territory of the Lausitz chapter, which opened at Görlitz in 1902, held 78 Society members in 1900, 60 percent of whom belonged to no chapter; the nearest chapter town was on average 83 km away before the opening and 19 km after it. The design treats each opening as a dated event for all places in the chapter's territory.

3. Conceptual Framework

The opening of a chapter changed whether the engineers, designers, and owners of firms in a region met regularly to discuss technical problems. Across countries, engineering associations offered "venues for regular meetings where informational exchange occurred formally in the presentation and discussion of papers and informally in conversations and personal relationships" (Beatty and Solares, 2025). Much of the knowledge that mattered in engineering design was of the kind exchanged at such meetings: how a machine behaved in use and which designs had failed (Vincenti, 1990; Ferguson, 1992). Such knowledge often stayed inside firms unless engineers met.

In the Cleveland iron district of northern England between 1850 and 1875, the owners and engineers of competing ironworks read papers on the design and operating results of their blast furnaces before engineering societies, among them the Cleveland Institution of Engineers, and visited each other's works; furnace height and blast temperature rose across the district as a result (Allen, 1983). In other settings, too, ideas have traveled along personal and professional ties (Becker et al., 2020a; Becker et al., 2024).

An opening could raise invention through two channels: by attracting engineers, firms,

[5]Annual pay of industrial workers from Desai (1968) and Bry (1960).

or new fields of activity to its territory, or by raising the output of the engineers and firms already there. The historical evidence favors the second channel. A chapter created no jobs or contracts, it was founded by members who already lived in the region, and its papers and plant visits dealt with the problems of the local firms whose engineers attended.

The two channels are not exclusive. A chapter may have made its town more attractive to engineers and firms, and the density of engineers may have changed for other reasons. The data separate the channels only imperfectly: Society members are observed in 1900 and 1925, census engineers only in 1925.

The second channel implies five patterns, which the empirical sections examine:

(1) *Timing.* The effect builds up over time, because relations take years to form and new designs take years to develop (Section 6.1).

(2) *Industrial places.* The effect is larger in industrial places, where more engineers and firms could meet (Section 6.4).

(3) *Industrial fields.* The effect is larger in the industrial patent classes, the fields in which the members worked (Section 6.4).

(4) *Local knowledge.* New patents draw more on the earlier work of their place (Section 7).

(5) *No inflow.* If the chapter mainly raised the output of the engineers and firms already in place, neither engineers nor industry grows faster where a chapter opened, and a new supply of technicians without a chapter, such as a machine-building school, does not raise patenting (Section 6.4). An inflow would not rule out an effect of the meetings, but it would mean that part of the effect cannot be attributed to forming a professional organization.

4. Data

The data place the chapters, the Society members and the high-value patents on one geography, the statistical units of 1900, and follow them from 1877 to 1918. Appendix A describes

the construction of the data in detail.

4.1. The Society: Directories, Chapters, and Territories

I digitize the Society's membership directories (Mitgliederverzeichnisse) of 1900 and 1925 and the membership of each chapter from 1857 onwards (Appendix A.1). The directories list every member under the chapter with which he was registered, with name, title or position, employer, street, and town; members without a chapter are listed separately. The directory of 1900 has 15,513 entries, 13,465 of which can be placed in their town; that of 1925 has 27,176 entries.

The directories show who met in the chapters. Seven in ten entries of 1925 are salaried engineers, 8 percent are directors and top managers, 6 percent independent engineers and consultants, and 3 percent owners. Only 4.5 percent carry the diploma of a technical university (Dipl.-Ing.), and 4.4 percent its doctorate (Dr.-Ing.). Prominent members appear next to ordinary ones: in 1900 the Württemberg chapter listed Robert Bosch with his workshop for precision mechanics and Wilhelm Maybach as director of the Daimler motor company, the Berlin chapter Emil Rathenau as general director of AEG, and the Bavarian chapter "Diesel, Rud. Ing., München, Schackstr. 2." Diesel's engine patent 67207 is among the high-value patents, and in June 1897 he presented the engine to the Society's annual meeting in Kassel (Diesel and Schröter, 1897). The main estimates do not use such links; they count patents by place and year.

Society members were far fewer than all engineers: in 1925, their number equaled about one in five of the engineers counted by the census (0.20 to 0.23, depending on the geography). I therefore write Society members for the counts from the directories and census engineers for those from the census. The yearbooks of 1882–1925 give the founding date, the founding roll and the territory of each chapter. The Society drew the territories itself, each a town and the districts around it, and printed them as a map in the yearbook of 1925. I assign each statistical unit to the territory that covers the larger part of its area.

4.2. High-Value Patents

The outcome is the number of patents renewed for at least ten years (Streb et al., 2006). Of the patents granted by the Imperial Patent Office between 1877 and 1918, 39,360 were renewed for ten years or more in the data that Streb provided,[6] and 25,800 of these were held in Germany. Under the patent law of 1877, a patent cost 30 marks at grant, and from the second year an annual fee began at 50 marks and rose by 50 marks a year, so keeping a patent in force for ten years cost about 2,300 marks, about two and a half years of an industrial worker's earnings around 1900 (Bry, 1960).

The count therefore weights invention by the holder's own valuation, as in Donges et al. (2023) and Cinnirella et al. (2025). Each patent carries its grant year, one of 89 technology classes and the town of its holder; a patent is located where its holder resided, so the patents of a firm are located at its seat. Patents were granted about a year after the application; Diesel applied for patent 67207 in February 1892 and received it in February 1893. High-value patents rose from 233 in 1878 to 2,311 in 1913 and fell during the war; year effects absorb such common changes. By their titles, I group the classes into 50 industrial classes, the fields of the Second Industrial Revolution such as machinery and engines, electrical engineering, chemistry, metals, and transport, and 38 traditional classes, mostly of crafts and consumer goods; agriculture counts as industrial because most of its high-value patents were farm machinery. The remaining class, salt works, holds 9 high-value patents and counts as traditional (Table A2).

Two further sources measure patenting without the renewal filter and the content of patents. The register of all 306,070 patents granted between 1877 and 1918 (Griesshaber and Streb, 2026) gives the grant year, class, title, and holder of every patent. For the content of patents, I use the text embeddings that Google Patents publishes for German patents, described with the text evidence below.

[6] Streb et al. (2006) report 39,343 such patents.

4.3. Places, Censuses, and Controls

The unit of observation is the statistical unit of 1900, which I call a place: a district (Kreis) or its equivalent outside Prussia, such as a Bezirksamt in Bavaria or an Oberamt in Württemberg. Germany had 1,020 such units. The design uses the 493 places in the territories of the chapters that opened between 1880 and 1912. All of them have data from the census of 1882; the population density of 1900, which the specifications of Panel A of Table 2 hold fixed, exists for 469 of them. As a second geography, I use the districts of 1882 (Kreise) of Donges et al. (2023), 365 of which lie in the design; I refer to them as the Kreise of 1882.

Four occupational censuses measure industry and engineers. The census of 1882, taken from the published volumes and, for Prussia, from the ifo Prussian Economic History Database (Becker et al., 2014), precedes all openings in the design except the four of 1880 and 1882; it gives the industry and agriculture shares of the workforce and the number of machine and instrument workers. The censuses of 1895 and 1907 count the workers of the machine, instrument, and electrical industries. Only the census of 1925 counts people by occupation; its row 70 counts employed engineers, architects, and builders. Industry is thus observed before the chapters opened, engineers only afterwards. I map all four censuses, which report 943 to 1,414 districts, to the statistical units of 1900 (Appendix A.2).

Population in 1871 comes from GPOP (Roesel, 2023), which covers the territory of today's Germany (Appendix A.2). The railway kilometers of each place in 1871, 1880, 1885, 1890, 1900, and 1910 come from RShapes (Cederman et al., 2025), each carried forward to the next benchmark year. The locations of the 29 state machine-building schools (Maschinenbauschulen), 24 of which opened during the period, serve for a placebo test.

Descriptive statistics. High-value patents were rare and concentrated (Table 1): the median place held two high-value patents over 42 years, the mean was 24.9 and the largest place held 5,668. The places of the design were mostly agrarian in 1882, with a median agriculture share of 0.59 and a median industry share of 0.26.

Industry, Society members, and high-value patents were concentrated in the same places,

Table 1. Summary Statistics

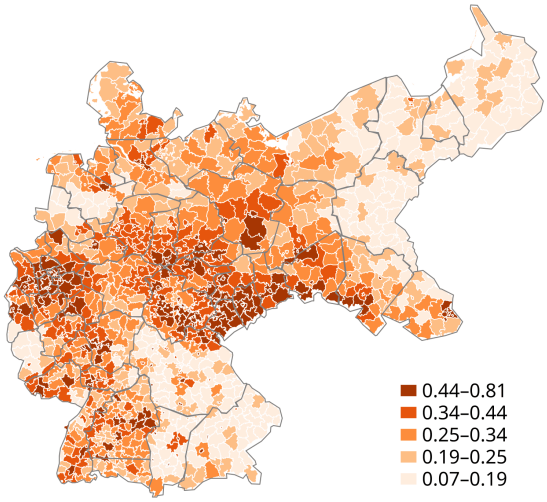
	Mean	Median	SD	Min	Max
Panel A. Statistical units of 1900, all of Germany (N = 1,020)					
High-value patents, 1877–1918	24.9	2	196.1	0	5,668
engineering classes	8.5	0	91.1	0	2,802
chemical classes	5.9	0	49.2	0	842
Society members, 1900	13.2	1	61.5	0	1,478
Population, 1871 (thousands)	28.7	24	44.6	0	1,029
Railway kilometers, 1885	32.4	28	27.5	0	226
Panel B. Statistical units of the design (N = 469–493)					
Industry share of the workforce, 1882	0.31	0.26	0.15	0.07	0.81
Agriculture share of the workforce, 1882	0.55	0.59	0.22	0.00	0.90
Machine and instrument workers per 1,000, 1882	5.8	3.6	8.3	0.8	94.4
Engineering-industry workers per 1,000, 1907	5.2	2.3	8.1	0.1	78.2
Census engineers, 1925	95	23	289	0	2,870
Society members, 1900	7.6	1	32.9	0	433
Panel C. Chapters					
Founding year (50 chapters)	1882	1878		1856	1924
Members registered with each chapter, 1900	297	259	266	61	1,612
Members registered with each chapter, 1925	549	382	631	85	4,352

Notes: High-value patents: patents renewed for ten years or more, located by the town of the holder (Streb et al., 2006); engineering and chemical classes are two subsets of the 50 industrial classes. Society members: entries of the 1900 directory placed in the unit. Population: GPOP (Roesel, 2023), which covers the territory of today's Germany (zero outside it); railways: RShapes (Cederman et al., 2025). Panel B: the 1882 variables for the 469 places of Table 2, Panel A; engineering-industry workers: machine, instrument, and electrical industries in the census of 1907; census engineers: row 70 of the census of 1925 (engineers, architects, and builders). Panel C: the 50 chapters founded in 1856–1924; members registered with each chapter as counted by the Society, for the 40 of these chapters that existed in 1900 and for the 49 of 1925.

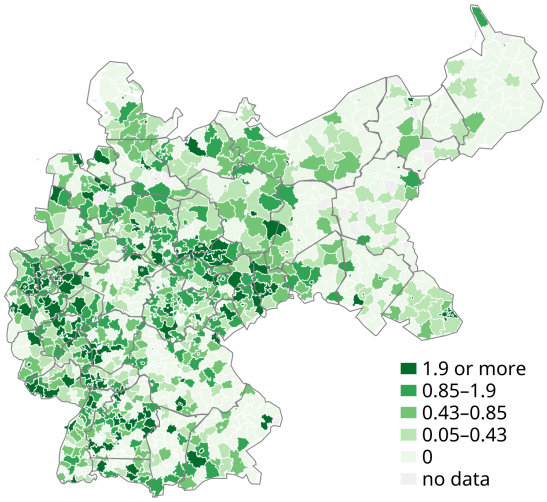
the Rhineland and the Ruhr, Saxony, Berlin, and the southwest, and all three were sparse in the agrarian east (Figure 2). Within chapters, places with more Society members in 1900 also held more high-value patents in 1900–1918: across the 493 places of the design, 1 percent more Society members are associated with 0.65 percent more high-value patents, with chapter effects and the population of 1871 held fixed (Figure 3). Chapters opened where engineers already worked, and those places might have patented more without a chapter, so this correlation is not an effect of the chapters. Because most place-years have no high-value patent, the outcome is $\ln(1 + \text{high-value patents})$; Appendix B.3 reports count models.

Figure 2. Industry in 1882, Society Members, and High-Value Patents per 10,000 Inhabitants

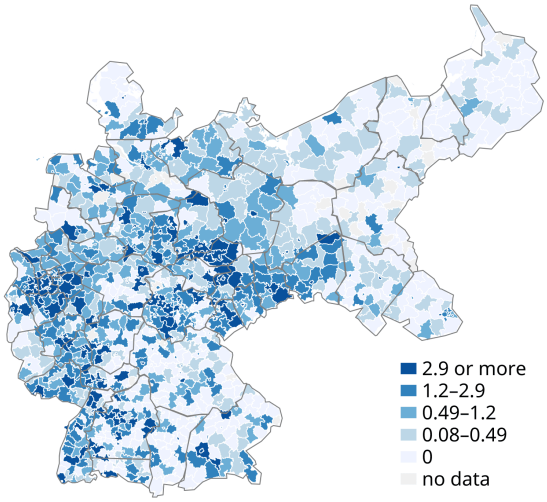
(a) Industry share of the workforce, 1882



(b) Society members per 10,000 inhabitants, 1900

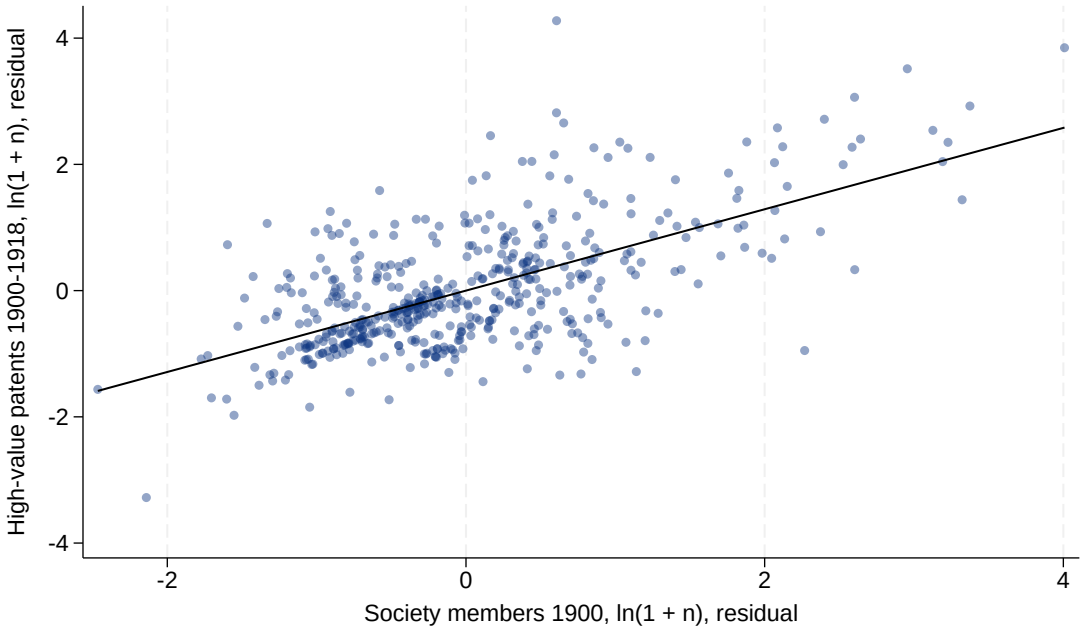


(c) High-value patents per 10,000 inhabitants, 1877–1918



Notes: Statistical units of 1900. Panel (a): share of industry in the workforce in the occupational census of 1882 (Appendix A.2). Panel (b): entries of the 1900 directory per 10,000 inhabitants of 1900. Panel (c): high-value patents granted 1877–1918 per 10,000 inhabitants. Lines: chapter territories at their 1925 shape. Light gray: no data (no population: 34 of 1,022 map units). Population of 1900 from the district data of Donges et al. (2023).

Figure 3. Society Members in 1900 and High-Value Patents of 1900–1918 within Chapters



Notes: One dot per place: ln(1 + high-value patents) of 1900–1918 against ln(1 + Society members) of 1900, both as residuals on chapter effects and the log population of 1871; the 493 places of the design. Line: slope of the regression with these controls, 0.645 (robust standard error 0.042). Society members: entries of the 1900 directory located in the place; high-value patents: Streb et al. (2006). Population of 1871: census of 1871 for Prussia, elsewhere the census population of 1882 projected back with the growth of 1882–1895 (Appendix A.2). Figure A1 shows the two variables without controls for all places.

5. Empirical Strategy

Treatment. A place is treated from the year in which the chapter whose territory covers it was founded. Let G_i denote this year for place i, and let $D_{it} = \mathbf{1}[t \geq G_i]$; I call the places of a territory whose chapter has opened chapter places, and the others not-yet-treated places. The 22 chapters of 1880–1912 opened in 16 founding years, and their territories hold 493 places, observed every year from 1877 to 1918, or 20,706 place-years. The places of the 25 chapters founded by 1877 were treated throughout and are not part of the design. Every place in the design received its chapter by 1912, so there are no never-treated places; the comparison group of a cohort is the places whose chapter opened later. The cohorts differ in size: the two chapters of 1892, Franconia–Upper Palatinate and West Prussia, cover almost a quarter of the places, and the Bochum chapter of 1891 covers four statistical units.

5.1. Estimator

With staggered openings and effects that grow over time, a two-way fixed effects regression on D_{it} would use places whose chapter had already opened as controls for places treated later, and some treated place-years would enter with negative weights (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Baker et al., 2022). I therefore use the doubly robust estimator of Callaway and Sant'Anna (2021), which builds on Sant'Anna and Zhao (2020) and compares each founding cohort only with not-yet-treated places.

For cohort g and calendar year $t \geq g$, the estimator compares the change in the outcome since the year before founding with the same change among the places whose chapter has not opened by t:

$$ATT(g,t) \;=\; E\big[\, Y_{it} - Y_{i,g-1} \,\big|\, G_i = g \,\big] \;-\; E_{\omega}\big[\, Y_{it} - Y_{i,g-1} \,\big|\, G_i > t \,\big], \tag{1}$$

where $Y_{it} = \ln(1 + P_{it})$, P_{it} is the number of high-value patents held in place i and granted in year t, and the subscript ω indicates that the comparison places are reweighted to match the covariates of the cohort; an outcome regression on the same covariates makes the estimator consistent if either part is correctly specified. The main estimate θ averages the $ATT(g,t)$ over all post-founding cells with at least one comparison place, with weights proportional to cohort size, so that cohorts with more years between their opening and 1912 carry more weight; the cohort of 1912 has no comparison places after its opening and serves only as a comparison group.

The event study $\theta(e)$ averages the cells by years since founding, $e = t - g$, again weighted by cohort size, for $e = -16, \ldots, 15$ (Figure 5 draws -15 to 15), with the year before founding as the base year. The covariates are railway kilometers and the population of 1871 interacted with a trend. The estimator uses their values in the base year $g - 1$, so that the second amounts to the population of 1871, and conditioning on them lets places with different railway access and size follow different trends. Later specifications add the industrial structure of

1882, the population density of 1900, the structure of 1895, and the stock of Society members in 1900. Standard errors are clustered by place; clustered by the 22 chapter territories instead, the estimate on all 493 places, 3.8 percent, has a standard error of 1.4 percentage points, and a wild cluster bootstrap gives a 95 percent interval of 1.0 to 6.6 percent.[7]

5.2. Identification and the Comparison Group

The comparison identifies the average effect of an opening on the places it covered under two assumptions: given the covariates, chapter places would have followed the same path as the reweighted not-yet-treated places, and places did not anticipate an opening. It also requires that an opening changed the patenting only of the places it covered; engineers from not-yet-treated places taking part in a neighboring chapter, or firms moving into a new territory, would affect the comparison places as well (Section 6.4). Differences in the level of patenting between early and late places drop out. The estimate would be biased if chapters opened where patenting was about to rise anyway (Section 6.3).

Chapter places and not-yet-treated places followed similar paths until their chapter opened (Figure 4). In raw means, panel (a), the two groups patent at similar levels before founding, a gap of -0.2 percent on average, and the chapter places pull away after it.[8] Relative to the year before founding, panel (b), the two groups move together over fifteen years and separate only after the opening.

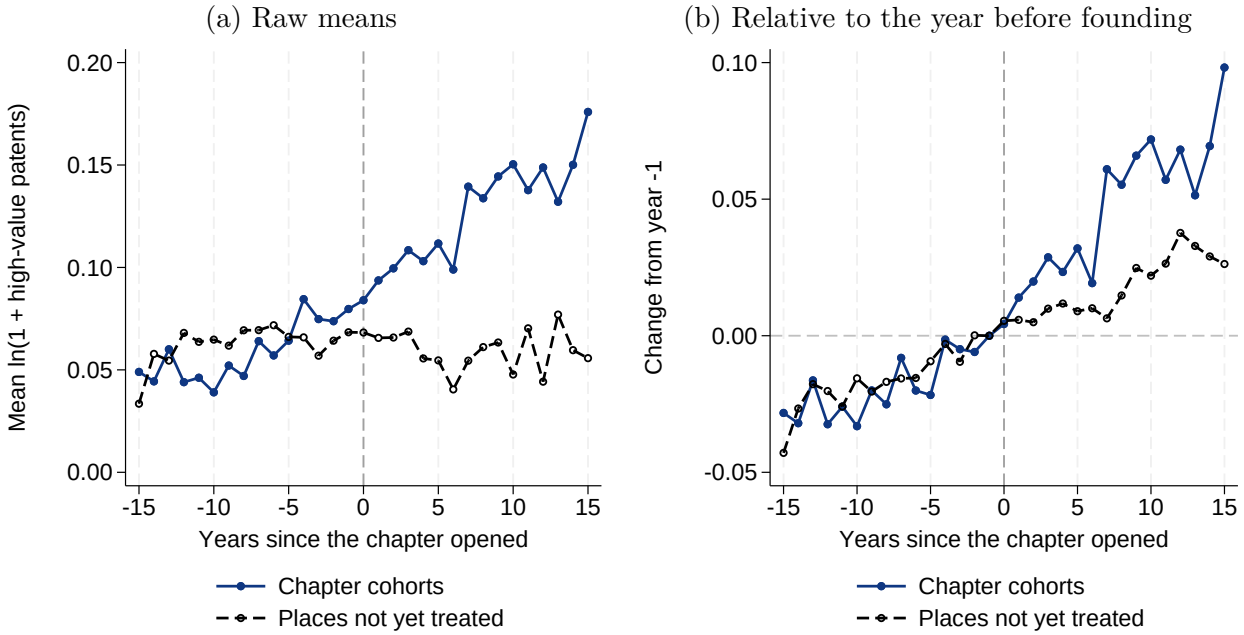
Because every place eventually received a chapter, the comparison group shrinks over calendar time: 72 places in 1903, 23 in 1904–07, 10 in 1908–11, and none after 1911. From 1904 on, the comparison places are those of the last two chapters, Bodensee (1908) and Mosel (1912), and from 1908 on those of the Mosel chapter alone.

I therefore also estimate designs that hold the comparison group fixed. The stacked

[7] A coefficient on $\ln(1+x)$ is not a percentage effect when the outcome can be zero, and its size depends on the units of the outcome (Chen and Roth, 2024); Appendix B.3 reports Poisson estimates, which measure percentage effects on the counts.

[8] The line of the chapter places rises before founding in part because the early cohorts enter the pre-founding years only from 1877, which changes its composition.

Figure 4. High-Value Patents of Chapter Places and of Places Not Yet Treated around Founding



Notes: Mean ln(1 + high-value patents) of the chapter cohorts by years since their chapter opened, and of the not-yet-treated places in the same calendar years; 493 places, 1877–1918. Panel (a): levels; cohorts enter the years before founding as the data allow, from 1877. Panel (b): every place relative to its own value in the year before founding. Dark blue: chapter cohorts; black, dashed: not-yet-treated places.

design of Cengiz et al. (2019) compares each cohort, over the years −5 to +8 around its founding, with clean controls: places treated more than eight years later and the 34 places outside the design whose territory had no chapter by 1918. Two windows end the sample in 1903 or 1907, so that the later cohorts serve as untreated controls throughout; for the window to 1903, I also estimate the imputation estimator of Borusyak et al. (2024) and a regression with cohort-by-period effects (Wooldridge, 2025). These designs measure mainly the first decade after an opening (Appendix B.2).

6. Results

6.1. Main Estimates

The opening of a chapter raised invention in its territory. Places in the territory of a new chapter obtained 4.8 percent more high-value patents a year than places whose chapter had

not yet opened, with railway kilometers, the population of 1871, and the population density of 1900 held fixed (Table 2, Panel A, column 1; $p < 0.01$). The density exists for 469 of the 493 places; on all 493 places without it, the estimate is 3.8 percent (Figure 5(b); Table B2, first row; $p < 0.01$), and the event studies, the estimates by industrial third and patent class, and the tests for spillovers use this full sample.[9] The effect grows over time: it averages 3.0 percent per year over the first 16 years after an opening, which adds up to about half of a place's annual high-value patents. Controlling for prior industrialization yields point estimates between 3.0 and 5.5 percent (Table 2, Panel A, columns 2, 4, and 5; Table 3). The results for the later chapters are similar: for the twelve chapters opened after the census of 1895 and the seven opened after the directory of 1900, controlling for the industry of 1895 or the stock of Society members of 1900 yields estimates about as large or larger (Panels B and C; Section 6.2).

Figure 5 shows the dynamics of the effect. Panel (a) uses the estimator of de Chaisemartin and D'Haultfœuille (2026), which compares the change in the outcome of the places whose chapter opened with the same change among the not-yet-treated places, and tests for differences before founding in the same way. Before founding, the two groups moved in parallel: the placebos lie between -2.1 and 1.8 percent and are jointly insignificant ($p = 0.47$). After founding, the effect builds up, in line with the first pattern of the conceptual framework, to between 4.6 and 9.3 percent in years 7 to 15; averaged over the years after founding, it is 4.5 percent ($p < 0.01$). This timing fits the years it takes to develop a design and the year between the application for a patent and its grant.

Panel (b) shows the event study of the doubly robust estimator of Callaway and Sant'Anna (2021), which in addition reweights the not-yet-treated places to match the covariates of each cohort. Its pre-founding coefficients average 0.6 percent and are jointly insignificant ($p = 0.42$); after founding, the coefficients reach 3 to 6.5 percent in years 8 to 15. Both estimators compare the chapter places with the same not-yet-treated places, so

[9] In the text, I report coefficients as percent effects and their significance as p-values; standard errors, clustered by place, are in the tables.

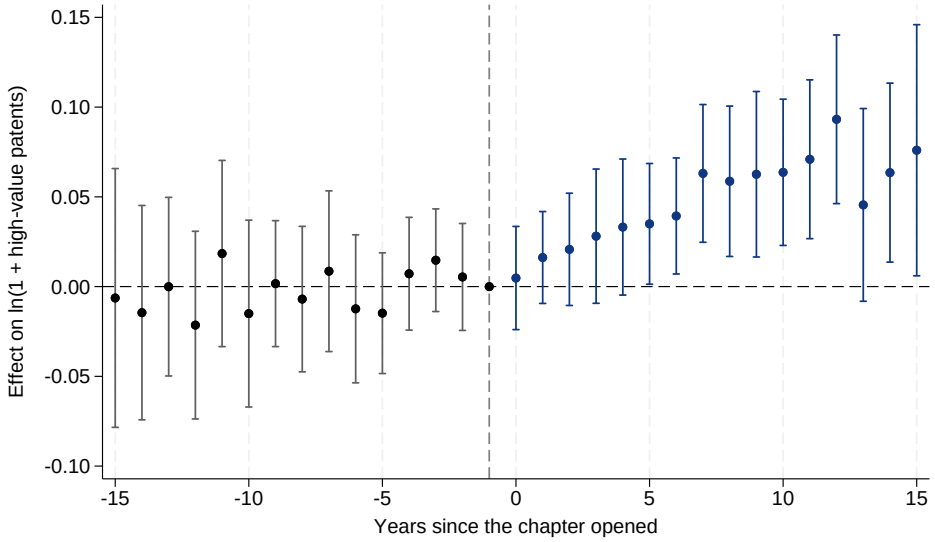
Table 2. High-Value Patents after a Chapter Opened: Prior Industrialization and the Stock of Society Members Held Fixed

	Dependent variable: ln(1 + high-value patents)				
	(1)	(2)	(3)	(4)	(5)
Panel A. All 22 chapters, 1880–1912					
Chapter opened	0.048***	0.030**	0.042***	0.042***	0.033*
	(0.017)	(0.015)	(0.015)	(0.016)	(0.017)
Railway km, population 1871 × trend	Yes	Yes	Yes	Yes	Yes
Industry and agriculture shares 1882	No	Yes	No	Yes	Yes
Machine workers and population 1882	No	No	No	No	Yes
Estimator	CS	CS	dCDH	dCDH	dCDH
Places	469	469	469	469	469
Panel B. Twelve chapters opened after the 1895 census, 1897–1912					
Chapter opened	0.071**	0.141***	0.138***	0.180***	
	(0.030)	(0.030)	(0.048)	(0.035)	
Railway km, population 1871 × trend	Yes	Yes	Yes	Yes	
Industry share and engineering industries 1895	No	Yes	No	Yes	
Industry share and machine workers 1882	No	No	Yes	Yes	
Estimator	CS	CS	CS	CS	
Places	226	226	226	226	
Panel C. Seven chapters opened after the 1900 directory, 1902–1912					
Chapter opened	0.098**	0.100**	0.086**	0.086**	0.084***
	(0.039)	(0.039)	(0.038)	(0.033)	(0.031)
Railway km, population 1871 × trend	Yes	Yes	Yes	Yes	Yes
Society members per 1,000, 1900	No	Yes	Yes	Yes	Yes
Members in no chapter per 1,000, 1900	No	No	Yes	Yes	Yes
Industry share, machine workers, and population 1882	No	No	No	No	Yes
Estimator	CS	CS	CS	dCDH	Imputation
Places	100	99	99	99	99

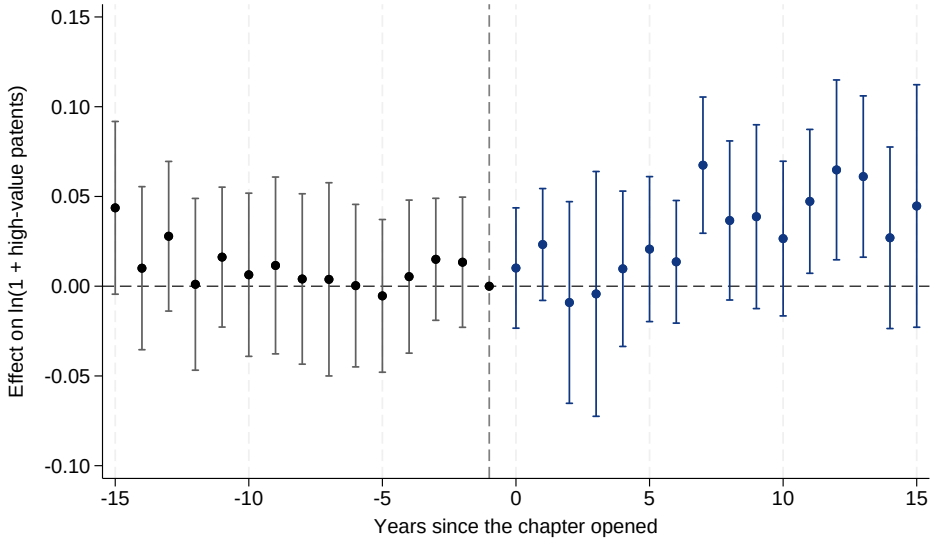
Notes: Place-year observations, 1877–1918. The columns of Panel A also hold the population density of 1900 fixed, which exists for 469 of the 493 places (Table B2 reports the estimate for all 493 places without it, 0.038 (0.014)). CS: estimator of Callaway and Sant'Anna (2021), doubly robust, with the places whose chapter has not yet opened as the comparison group; the listed covariates enter the reweighting and the outcome regression. Appendix D.1 reports CS with the full 1882 set, which rests on the 10 to 23 comparison places of the cells of 1904–11 and is imprecise. dCDH: estimator of de Chaisemartin and D'Haultfœuille (2026) with the not-yet-treated places as controls and the covariates of the column, each time-invariant covariate interacted with a linear trend; the estimate is the average effect over years 0 to 15 after founding (Panel C: years 0 to 8), the design of Figure 5(a). Imputation (Panel C, column 5): estimator of Borusyak et al. (2024), with each covariate interacted with a linear trend and fitted on the not-yet-treated place-years, on the years to 1911, after which no place is untreated (Table 3 reports it for Panel A). Panels B and C: the chapters opened after the census of 1895 and after the directory of 1900, each against its own not-yet-treated places, from 1904 on those of the Bodensee and Mosel chapters; only the comparison within each panel is informative. Panel C covers the years 1895–1918, column (5) the years 1895–1911. Engineering industries: workers in the machine, instrument, and electrical industries; Society members and members in no chapter per 1,000 inhabitants, in logs. Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 5. Event Studies of High-Value Patents around the Opening of a Chapter

(a) de Chaisemartin and D'Haultfœuille



(b) Callaway and Sant'Anna



Notes: Coefficients on ln(1+high-value patents) by years since the chapter opened, with 95 percent confidence intervals; 493 places, 1877–1918; comparison group: places whose chapter has not yet opened; railway kilometers and the 1871 population × trend; standard errors clustered by place. All coefficients are relative to the year before founding (dashed line), which is zero by construction; from the year of founding (year 0) on in dark blue. Panel (a): estimator of de Chaisemartin and D'Haultfœuille (2026); the coefficient for year $e \geq 0$ compares the change in the outcome from the year before founding to year e with the same change among the not-yet-treated places, and the coefficient for year $e < -1$ is the placebo that compares the change from year e to the year before founding in the same way; average effect after founding 0.045 (0.013); joint tests of the placebos, $p = 0.47$, and of the effects, $p = 0.008$. Panel (b): doubly robust estimator of Callaway and Sant'Anna (2021); event-study coefficients $\theta(e)$, the cohort-size-weighted averages of the $ATT(g,t)$ of Equation (1); average effect over all post-founding cells 0.038 (0.014), the estimate on all 493 places; the coefficients for years -10 to -2 average 0.006 (0.018); joint test of the coefficients before founding (years -16 to -2), $p = 0.42$.

the path does not depend on the estimator, and the estimate of 3.8 percent on all places is the smaller of the two. Figure B1 shows the event study of the imputation estimator.

The comparison group. The comparison group of the main design shrinks over time (Section 5.2; Table B1). The designs that hold it fixed cover mainly the first decade after an opening: the stacked design with clean controls, the windows that end in 1903 and 1907, and the imputation and cohort-by-period estimators on the window to 1903 put the effect over this decade at about 1 percent (Appendix B.2), statistically indistinguishable from the 2.3 percent of the main design over the same years. Adding the 34 never-treated places to the comparison group gives 1.3 percent, and conventional two-way fixed effects give 1.5 percent ($p < 0.05$). The designs with a fixed comparison group do not reach beyond the first decade, when the effect grows; Appendix B.2 reports estimates of the effect by calendar period and cohort.

6.2. Prior Industrialization and the Stock of Society Members

The main concern is selection: chapters may have opened in places that were already industrializing, where patenting would have risen even without a chapter. Panel A of Table 2 therefore holds the industrial structure of each place fixed, step by step, with railway kilometers, the population of 1871, and the population density of 1900 held fixed throughout. On the 469 places for which the density exists, the estimate is 4.8 percent (column 1), and adding the industry and agriculture shares of 1882 gives 3.0 percent (column 2). Adding machine workers and the population of 1882 as well leaves the estimator of Callaway and Sant'Anna (2021) with too little support: with seven covariates, its reweighting rests on the 10 to 23 comparison places of the cells of 1904–11, and the estimate becomes imprecise (Appendix D.1). The estimator of de Chaisemartin and D'Haultfœuille (2026) instead estimates the trends of the covariates from all not-yet-treated place-years rather than cell by cell. Columns (3) to (5) repeat the walk with this estimator: 4.2 percent without the 1882 variables, 4.2 percent with the shares and 3.3 percent with the full set ($p < 0.01$, $p < 0.01$

Table 3. Estimates with the 1882 Structure Held Fixed: Imputation Estimator by Geography

	Dependent variable: ln(1 + high-value patents)	
Covariates, each × trend	Statistical units of 1900	Kreise of 1882
Railways, population 1871	0.059***	0.048***
	(0.011)	(0.015)
+ population density 1900	0.057***	0.048***
	(0.011)	(0.015)
+ industry and agriculture shares 1882	0.055***	0.048***
	(0.012)	(0.014)
+ machine workers and population 1882	0.050***	0.049***
	(0.012)	(0.014)
Places	469	365
Place-years, 1877–1911	16,415	12,775

Notes: Imputation estimator of Borusyak et al. (2024) on ln(1 + high-value patents), 1877–1911: place and year effects and the covariates, each interacted with a linear trend, are fitted on the not-yet-treated place-years; the untreated outcome of every treated place-year is imputed from this fit, and the estimate is the average difference between the actual and the imputed outcome over all treated place-years. Rows add covariates cumulatively, so that places with a different population density in 1900 or a different industrial structure in 1882 may follow different linear trends. Columns: the 469 statistical units of 1900 of Table 2, Panel A, and the 365 Kreise of 1882 of Donges et al. (2023) with 1882 census data. The sample ends in 1911, after which no place is untreated and the year effects are not identified. Standard errors clustered by place (Kreise: by Kreis) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

and $p = 0.05$). The point estimates of all five columns lie between 3.0 and 4.8 percent.[10]

Panel B uses the census of 1895, which preceded twelve of the 22 openings. Within this panel, holding the industry share and the engineering industries of 1895 fixed raises the estimate from 7.1 to 14.1 percent; the industrial structure of 1882 gives 13.8 percent and both sets together 18.0 percent ($p < 0.01$ in all three). The change in the engineering industries and in the industry share between 1882 and 1895 did not predict whether or when a chapter opened (Appendix C.2).

Panel C holds the stock of Society members fixed. The seven chapters that opened after the directory of 1900, from Lausitz in 1902 to Mosel in 1912, allow this test, because the directory counts the Society members in each of their 100 places before the chapter opened. Territories without a chapter in 1900 had about as many members per head as territories

[10]With the sample ending in 1907, so that the cohorts of 1908 and 1912 form a fixed comparison group, the estimates lie between −1.0 and 1.8 percent, and the 1882 structure moves them only within their standard errors (Appendix B.2).

with one, 0.091 against 0.083 per 1,000 inhabitants. The estimate barely moves when these members are held fixed: 9.8 percent without them, 10.0 percent with the members per 1,000 inhabitants, 8.6 percent with the members in no chapter as well, 8.6 percent with the same covariates in the estimator of de Chaisemartin and D'Haultfœuille (2026), and 8.4 percent with the 1882 structure added through the imputation estimator ($p < 0.05$ in all; Appendix D.1 reports the doubly robust estimator with the full set, which is imprecise).

The imputation estimator of Borusyak et al. (2024), which lets places with a different industrial structure follow different linear trends, gives 5.0 to 5.9 percent on the statistical units and 4.8 to 4.9 percent on the second geography, the Kreise of 1882, each estimate significant at the 1 percent level (Table 3). Overall, controlling for prior industrialization yields estimates between 3.0 and 5.5 percent in the full sample, and controlling for the industry of 1895 or the stock of Society members yields estimates about as large or larger for the later chapters.

6.3. Pre-Trends and Selection into Founding

The raw means of Figure 4 show no divergence before founding that is significant at the 5 percent level (Section 5.2). Net of place effects, year effects, and controls, all estimated on the not-yet-treated place-years, the chapter places show at most a slight upward drift before founding; extrapolated, it would account for about half a percentage point of the average effect over years 0 to 9 (Appendix C.1). With the 1882 structure held fixed, the residuals are close to zero before founding and rise from about year 3 after it (Figure C1).[11]

The level of patenting and the distance to the nearest chapter predicted an opening; recent growth did not. In a discrete-time hazard model of the year in which a place becomes a chapter town (17 openings between 1883 and 1914), a standard deviation more patenting in the previous three years raises the annual probability of an opening by 1.72 per 1,000 places, against a base rate of 0.53, and a standard deviation more distance to the nearest

[11] Appendix C.1 reports the bounds of Rambachan and Roth (2023), which allow for violations of parallel trends that a pre-test may not detect (Roth, 2022).

chapter raises it by 0.47 ($p < 0.05$ for both), with railway kilometers held fixed. Faster growth of patenting did not raise it: the coefficient is negative for growth over three, five, and ten years, and significant at the 5 percent level for growth over three years (Table C1).

The place effects of the design absorb differences in levels. Early and late places also looked alike in 1882: the industry share was 0.29, 0.34, and 0.28 in the places of the chapters of the 1880s, 1890s, and 1900s, with no significant difference (Table C2).

The results do not rest on the chapter towns, where patenting grew most.[12] The chapter towns are 22 of the 493 places, and the stacked design without them gives 1.3 percent ($p < 0.1$), against 1.5 percent with them.

6.4. Inflows, Spillovers, and Heterogeneity

Engineers, new schools, and rings around chapter towns. The chapters raised invention without drawing engineers or industry into their territories. Where a chapter opened between 1900 and 1925, the number of Society members per inhabitant grew by an insignificant 4 log points more than elsewhere, and the number of members in no chapter by 2 log points; workers in the engineering industries and the industry share did not grow significantly faster either (Table D1).[13] Founding rolls rose over time even as chapters opened in smaller towns (Section 2.3).

More engineering capacity alone did not show the same pattern. The 24 state machine-building schools that opened during the period trained technicians but brought no chapter, and places 10 to 50 km from a new school did not gain significantly more high-value patents than places 50 to 100 km away: the imputation estimator gives 2.8 percent per year ($p = 0.14$). Around chapter towns, the same ring design shows the late rise of the main estimates: with the imputation estimator, the effect is 0.8 percent in years 0 to 7 and 4.7 percent in years 8 to 15 ($p < 0.01$), whereas around the schools it is 2.3 and 1.8 percent, without a later

[12] Compared with the places of the same territory more than 40 km away, a chapter town gained 38 log points after founding; the gain began before founding (Appendix C.3).

[13] The intervals are wide: for Society members, the 95 percent interval runs from −13 to 20 log points.

rise (Figure D1).[14]

Spillovers between places. The comparison assumes that an opening did not change the patenting of the not-yet-treated places. Two departures are plausible, with opposite biases. Engineers from neighboring places could take part in a new chapter; before the seven openings after 1900, 62 percent of the members in the future territories were registered with the nearest existing chapter, on average 54 km away. Such spillovers raise patenting in the comparison places and bias the estimate toward zero. Firms and inventors could instead move into a new territory, which lowers patenting in the comparison places and biases the estimate upward. The evidence leans toward the second bias. Dropping the 33 places within 25 km of a place whose chapter had opened earlier changes the estimate on all places only from 3.8 to 3.7 percent ($p < 0.05$), but dropping the 92 places within 50 km lowers it to 2.6 percent ($p = 0.10$). In two-way fixed effects regressions, not-yet-treated places with an opened place within 100 km held 2 to 3 percent fewer high-value patents, with a gap about as large at 50 to 100 km as within 25 km, and with these distance bands the coefficient of an own opening falls from 1.5 to 0.1 percent (Appendix D.5). Because the gap barely changes with distance, it may reflect a slower rise of the late cohorts rather than displacement, but the estimates do not rule out that part of the effect was patenting drawn from neighboring places.

Heterogeneity. The effect was largest where more engineers and firms could meet, in line with the second pattern of the conceptual framework. Estimated separately within thirds of the 1882 industry share, each against its own not-yet-treated places, the effect is 10.5 percent in the most industrial third ($p < 0.05$) and 5.5 percent in the middle third ($p < 0.01$), against -2.6 percent in the least industrial third ($p < 0.05$; Table E1). The negative estimate in the least industrial third may reflect some displacement within territories (Appendices C.3 and D.5).

[14] Over all years after the opening, the two kinds of ring differ by less than the standard error of either (Table D2).

The effect was concentrated in the industrial classes, in line with the third pattern. Table E3 defines the traditional classes in three ways: as in the broad split of Table A2; as crafts, consumer goods, and agriculture; and as crafts and consumer goods alone, fields that were largely artisanal and in which few engineers worked. High-value patents in the industrial classes rose by 3.4, 3.3 and 3.6 percent ($p < 0.01$ in all three), and those in the traditional classes by 1.2, 1.3 and 0.7 percent, none of them significant. In crafts and consumer goods, from bakery and clothing to toys and clocks, the rise is close to zero (standard error 0.9 percentage points), and the difference from the industrial classes, 2.9 percentage points, is significant at the 5 percent level.[15] The number of classes in which a place held a high-value patent rose by 3.1 percent ($p < 0.05$; Table E2).

7. Evidence from Patent Texts

If a chapter led the engineers of a place to learn from each other, the patents of the place should draw more on its own earlier work after the opening, the fourth pattern of the conceptual framework. The texts of the patents support this: after an opening, the high-value patents of chapter places became more similar to the earlier patents of their own place.

7.1. Measures

Google Patents publishes a text embedding for almost every German patent of the period: a list of 64 numbers that a model trained to predict the technology classes of patents computes from the patent's text (for German patents, from Google's English machine translation) (Abi Younes and de Rassenfosse, 2024). Patents with similar texts have similar embeddings; the embeddings cover 94.9 percent of the 306,070 patents in the register (Appendix A.4). Because the model was trained to predict modern patent classes, two patents of the same class are similar in part by construction.

[15] In the two broader definitions, the difference between the traditional and the industrial classes is not significant ($p = 0.15$ and $p = 0.13$).

The similarity of two patents is the cosine of the angle between their embeddings, multiplied by 100; it does not depend on the length of the texts. For each patent, I compute its mean similarity to the patents granted in the five preceding years, as in Arts et al. (2021) and, with the sum in place of the mean, Kelly et al. (2021), in two groups: the patents of its own place, of any class, and the patents of the same class in other chapter territories. The local premium is the difference between the two means.

Because the comparison group shares the patent's class and the local group does not, the premium is negative on average (-7.1 among the high-value patents of the 22 territories), and the regressions hold the class and year of each patent fixed. Pairs of patents with the same holder and pairs of a patent and its addition patent (Zusatzpatent) are excluded, so that the premium does not reflect holders building on their own patents.

The measure ranks pairs of patents as expected: an addition patent is far more similar to its parent patent than two patents of the same class are to each other, and these are more similar than two random patents; the similarity of Rudolf Diesel's patent 82168 of 1895 to his patent 67207, to which its claim refers, lies above the 99.9th percentile of random pairs (Appendix A.4).

7.2. Results

After an opening, the high-value patents of chapter places became more similar to the earlier patents of their own place. Table 4 reports patent-level regressions with place effects, class-by-year effects, railway kilometers, the population of 1871 interacted with a trend and the log number of local comparison patents. In column (1), the local premium of high-value patents rose by 1.43 points after founding ($p < 0.01$). A high-value patent's mean similarity to the earlier patents of its place is about 50, so high-value patents became about 3 percent more similar to the earlier patents of their own place, relative to patents of the same class elsewhere: a rise of 0.2 standard deviations of the premium, or 4 percent of the gap between two random patents (48) and Diesel's patent 82168 and the patent its claim refers to (81).

Permuting the founding years across territories 500 times gives $p = 0.008$, and among the high-value patents of all places the estimate is 1.17 ($p < 0.05$; Table E4).

The rise is confined to the place itself: similarity to the other places of the same territory, to foreign patents and to patents of other territories, which serve as placebo outcomes, did not change. It is also concentrated in the high-value patents; among all patents in the register, the estimate is a quarter as large (column 2). Part of this difference reflects which patents were renewed: after an opening, patents with a higher local premium were renewed more often than before, and this selection accounts for about 40 percent of the estimate for high-value patents (Appendix E.2).

Robustness of the text evidence. The rise in the local premium holds across samples and designs. Excluding pairs whose holders share a name, such as Siemens & Halske and

Table 4. Patent Texts after a Chapter Opened

	Places of the 22 chapters		
	(1) High-value patents	(2) All patents	(3) Mean of (1)
Local premium	1.426***	0.355	−7.10
	(0.518)	(0.283)	
Same territory, other places	−0.104	−0.164	49.63
	(0.446)	(0.226)	
Foreign patents (placebo)	0.314	0.057	50.10
	(0.383)	(0.167)	
Other territories (placebo)	0.314	0.079	48.61
	(0.345)	(0.148)	
Patents	5,365	45,512	
Places	201	300	

Notes: Patent-level regressions of each outcome on an indicator for a patent granted in a place whose chapter had opened, with place effects, class × year effects, railway kilometers, and the 1871 population × trend; patents of the places of the 22 chapters of the design. Column (1): high-value patents; column (2): all patents in the register; column (3): mean of the outcome in the sample of column (1). Patents and places: a patent enters only if its place had a patent in the five preceding years, which the local premium requires; fewer places have such a high-value patent (201 of the 493) than such a patent of any kind (300). Outcomes: cosine similarity of Google Patents embeddings ×100, averaged over the patents granted in the five preceding years; local premium: similarity to earlier patents of the same place minus similarity to earlier patents of the same class in other territories; same-holder pairs and pairs of a patent and its addition patent excluded; a coefficient of 1 is about 2 percent of the mean similarity of a high-value patent to the earlier patents of its place (about 50). The regressions also control for the log number of earlier local patents compared. Table E4 reports the local premium for the high-value patents of all places and further designs. Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Siemens-Schuckert, gives 1.18 ($p < 0.01$), dropping one territory at a time gives estimates between 1.06 and 1.79, and the stacked design with clean controls keeps the size of the estimate, 1.27, with a larger standard error (Appendix E.2).[16]

The local premium rose in the industrial classes, by 1.32 points ($p < 0.05$), and less and not significantly in the traditional classes, by 0.54 points (Table E4). Overall, the text evidence is consistent with the high-value patents of chapter places drawing more on earlier local work after an opening.[17]

8. The Long Run

Invention remained concentrated around the chapter towns long after they were established. I compare West German counties around chapter towns with counties around other cities, period by period from 1877 to 2023, with the patents granted to holders in the county from PatentCity (Bergeaud and Verluise, 2024), as $\ln(1 + \text{patents})$ with the county's population of 1871 held fixed.

This comparison can show persistence but not a causal effect of the chapters a century later: the engineers who met in the chapters of 1880–1912 were dead by the time most of these patents were granted. The question is whether the geography of invention that the chapters helped to shape outlasted them, that is, whether the regions around the chapter towns kept their lead once the firms, schools, and engineers that the chapters had connected were in place. The comparison is descriptive. The chapter towns were not chosen at random, and the controls hold fixed only what the county looked like in 1871 and where it lay; the placebo rings show what proximity to a city of the same size produced without a chapter.

The chapter rings compare the counties 10 to 50 km from a town whose chapter was founded by 1918 with the counties 50 to 100 km away, with the chapter town's own county

[16] With a five-year horizon, the stacked estimate is 0.68 and not significant, and in the stacked event study of all places the premium is already elevated in years -5 to -2 (Appendix E.2).

[17] The measure captures the similarity of patents rather than which patent drew on which, and it rests on machine translations.

dropped; the placebo rings draw the same rings around cities of 20,000 inhabitants in 1871 that had no chapter town within 12 km. All comparisons hold fixed the county's population and density in 1871, its distance to a city of 50,000 inhabitants and state effects (Appendix F describes the samples and the periods).

Around chapter towns, a gap in patents opened by the 1920s and has not closed since (Figure 6). In every period from 1919–32 to 2010–23, the counties near a chapter town hold 24 to 42 log points more patents than the counties farther away with the same population in 1871; the gap is significant at the 10 percent level in four of the six periods and at the 5 percent level in the two periods after 1990. Around cities without a chapter, the same rings show a negative gap in every period.

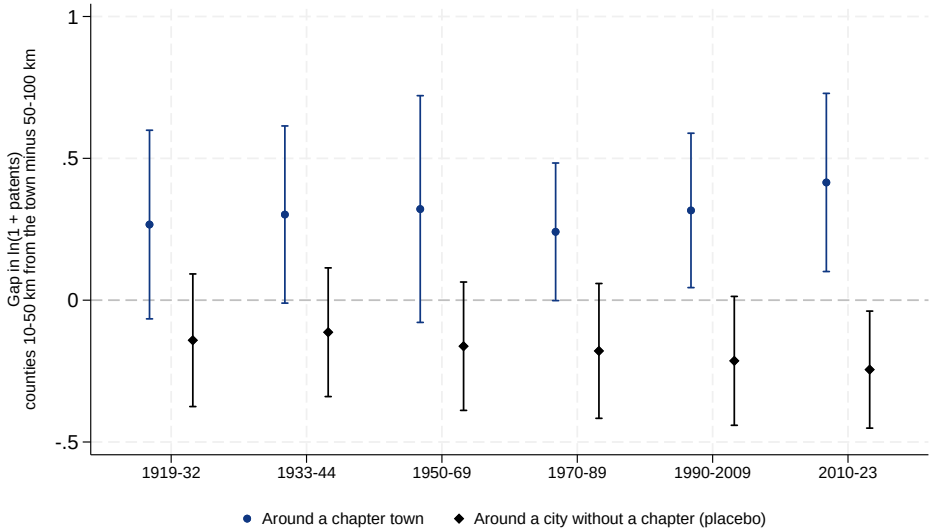
Pooled over the years after 1950, the chapter rings give a gap of 32 log points ($p < 0.05$) and the placebo rings one of -20 log points ($p < 0.05$). Counties whose nearest city of 20,000 inhabitants was a chapter town hold 36 log points more patents after 1950 than counties whose nearest city was not, with the size of the city, early patenting, and the 1882 industry held fixed ($p < 0.05$).[18]

After 1950, the gap is 51 log points around the towns whose chapter existed by 1877 ($p < 0.05$) and 24 log points around the West German towns of the chapters of the design ($p = 0.11$); the two do not differ significantly ($p = 0.20$; Appendix F). The gap also outlasted the Society's own upheavals: in 1933 the Society was integrated into the National Socialist organization of technology, and after the war it was refounded without its chapters in the East (Ludwig, 1981). Whether the chapters themselves, which the Society refounded in the West after 1945, or the firms and schools around them carried the lead across the century cannot be distinguished. The data show only that the areas around the towns that organized their engineers early still patent more than places farther away.

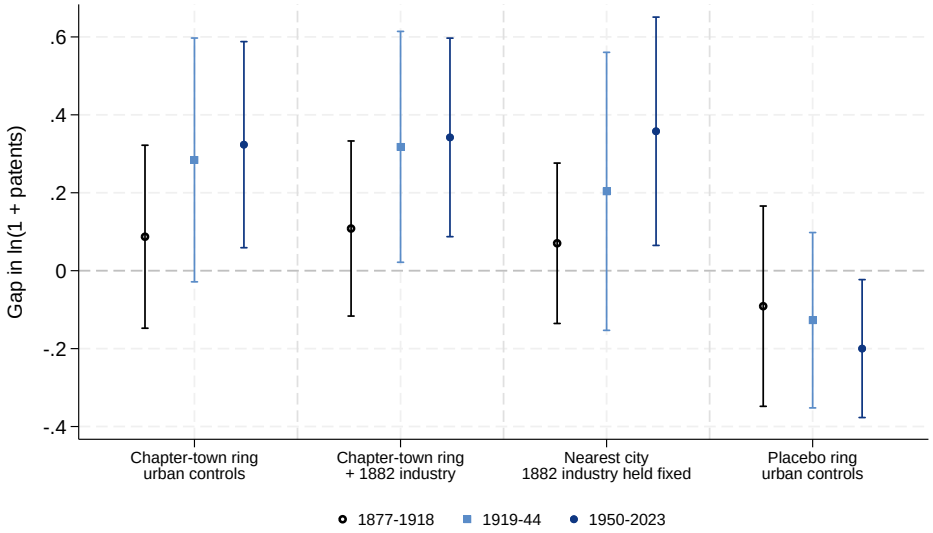
[18]Before 1918, the gap around chapter towns was 9 log points and not significant. County GDP shows no significant gap (Appendix F.2).

Figure 6. Patents around Chapter Towns and around Other Cities, 1877–2023

(a) By period: rings around chapter towns and around cities without a chapter



(b) Pooled by era: four comparisons



Notes: Gap in ln(1 + patents granted in the period), with 95 percent confidence intervals; the county's population of 1871 is held fixed in every regression, so that a gap of 0.25 is about 25 log points more patents than in a county of the same population in 1871. Panel (a): counties 10–50 km against 50–100 km from the town, one regression per period, around the 32 towns whose chapter was founded by 1918 (dark blue; 176 against 103 West German counties) and around cities of 20,000 inhabitants in 1871 without a chapter town within 12 km (black; 151 against 83 counties); the period 1877–1918, not shown, gives 0.09 (0.12) and −0.09 (0.13). Held fixed: the county's population and density in 1871, its distance to a city of 50,000, and state effects. Panel (b): the same outcome averaged within era (1877–1918 black, 1919–44 light blue, 1950–2023 dark blue) for the chapter rings, the chapter rings with the 1882 industry and early patents held fixed as well, the nearest-city comparison and the placebo rings. Standard errors clustered by town or city. Patents: PatentCity (Bergeaud and Verluise, 2024); population: GPOP (Roesel, 2023).

9. Conclusion

This paper examines whether the local chapters of the German Engineering Society, which brought together owners, salaried engineers, and teachers from different firms and schools, raised invention in the places they covered. Comparing places whose chapter had opened with not-yet-treated places, I find that places obtained 4.8 percent more high-value patents per year after an opening. The estimate is robust to controlling for the industrial structure of 1882. The effect grew over time; designs that hold the comparison group fixed, which cover mainly the first decade, put it at about 1 percent per year over those years.

The effect was largest in industrial places, and there is no evidence of a significant inflow of Society members, engineering workers, or industry. New machine-building schools, which added technicians but no chapter, did not significantly raise patenting nearby. After an opening, the high-value patents of chapter places became more similar to earlier patents from the same place. A century later, counties around the chapter towns still hold more patents than counties farther away with the same population in 1871.

Engineering societies spread across the industrial world (Beatty and Solares, 2025), and their successors still bring together the engineers of competing firms. The German case suggests that such organizations, by facilitating the exchange of knowledge among engineers, can shape where invention takes place and in which fields.

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Appendix

Appendix A documents the data. Appendix B counts the comparison places by cohort, defines the designs with a fixed comparison group and the count models, and collects the estimates for other samples and scales. Appendix C examines the path before founding and selection into founding, including the chapter towns. Appendix D holds prior industrialization and the stock of Society members fixed, tests for an inflow of engineers, compares rings around chapter towns and machine-building schools, and discusses spillovers between places. Appendix E covers heterogeneity and the patent texts, and Appendix F the long-run comparisons and the eighteenth-century economic societies.

Appendix A. Data

A.1. Directories, Chapters, and Territories

I digitized the membership directories of 1900 and 1925 from scans. Optical character recognition (Google Cloud) turned the pages into text, and a large language model (Claude, Anthropic) split the text into entries with name, title or position, employer, street, town, and chapter. I also used Claude to check the spelling, grammar, and formatting of the paper.

Before membership in a chapter became compulsory in 1911, a large part of the Society belonged to no chapter: 35 percent of the members in 1870 and 22 percent in 1900, against 6 percent in 1925. Of the 13,465 placed entries of the directory of 1900, 12,210 lie in the Reich.

Six chapters had breaks or were renamed: Westphalia (founded 1856, refounded 1864), Chemnitz (1866, 1886), Braunschweig (1872, 1879, 1883), Pomerania (Stettin until 1869), Frankfurt (Darmstadt until 1875) and Leipzig (Saxony until 1907); with alternative founding dates for these chapters, the estimate on all 493 places is 0.043 (0.015), against 0.038. Founding dates come from the yearbooks of 1882–1925, checked against the 47 chapters listed in the directory of 1910. The chapter of Alsace-Lorraine, founded in 1895 and ceded with its territory to France in 1919, is excluded throughout. Table A1 lists the 22 chapters of the design.

The territories are those of the map in the yearbook of 1925. Before 1911 their edges were less sharp, so the shape of 1925 may assign some border places to a chapter slightly too early or too late. The Posen chapter, founded in 1900, is missing from the map because the province was ceded after 1918.

Table A1. The 22 Chapters of the Design: Cohort, Chapter Town, Places, and Kreise of 1882

				Statistical units of 1900		Kreise of 1882
Cohort	Chapter	1925 yearbook	Chapter town	All	Density 1900	1882 data
1880	Ostpreußischer	35 Ostpreußen	Königsberg	34	33	32
1880	Schleswig-Holsteinischer	41 Schleswig-Holstein	Kiel	18	18	15
1882	Karlsruher	20 Karlsruhe	Karlsruhe	36	36	35
1882	Leipziger	23 Leipzig	Leipzig	10	10	8
1883	Hamburger	17 Hamburg	Hamburg	19	16	11
1888	Märkischer	26 Mark	Frankfurt (Oder)	21	20	17
1891	Bochumer	6 Bochum	Bochum	4	4	1
1892	Fränkisch-Oberpfälzischer	15 Franken-Oberpfalz	Nürnberg	91	89	67
1892	Westpreußischer	47 Westpreußen	Danzig	22	18	12
1894	Teutoburger	43 Teutoburg	Bielefeld	12	12	11
1897	Dresdener	12 Dresden	Dresden	9	9	7
1898	Bremer	9 Bremen	Bremen	30	28	20
1898	Mittelthüringer	30 Mittelthüringen	Erfurt	42	40	28
1899	Zwickauer	49 Zwickau	Zwickau	8	8	7
1900	Posener	not on the 1925 map	Posen	37	29	17
1902	Lausitzer	22 Lausitz	Görlitz	17	17	15
1903	Unterweser	45 Unterweser	Bremerhaven	11	10	7
1904	Augsburger	2 Augsburg	Augsburg	36	36	24
1904	Emscher	14 Emscher	Gelsenkirchen	2	2	1
1904	Rheingau	38 Rheingau	Mainz	11	11	8
1908	Bodensee	7 Bodensee	Friedrichshafen	13	13	13
1912	Mosel	31 Mosel	Trier	10	10	9
	Total			493	469	365

Notes: Chapters founded 1880–1912, in 16 founding years. 1925 yearbook: number and name on the map in the yearbook of 1925. Statistical units of 1900: the places of the design in the territory of the chapter, all 493, each with 1882 census data, and the 469 with the population density of 1900. Kreise of 1882: the districts of 1882 of Donges et al. (2023), the second geography of Tables 3 and E1, with 1882 census data (365 in all). The seven chapters of 1902–1912 cover 100 of the 493 places and 99 of the 469.

A.2. Censuses and Crosswalks

Crosswalks link the census districts to the district maps by name where possible and otherwise through the district seats, with about 90 percent of the population linked by name, and allocate each district to the statistical units of 1900 by the area and population density of their overlaps; the allocated totals equal those printed in the census volumes, and every place of the design receives census data. The censuses report 943 districts in 1882 (Statistik des Deutschen Reichs, N.F. vol. 2; for Prussia, Preußische Statistik vol. 76, via the ifo Prussian Economic History Database of Becker et al. (2014)), 1,010 in 1895 (N.F. vol. 109), 1,066 in 1907 (vol. 209) and 1,414 in 1925 (vols. 403–405, occupation row 70); the digitized tables of 1907 reproduce every total printed in the census volume.

Because the census of 1907 counts family helpers in agriculture more completely than that of 1882, levels of the industry share are compared only within one census, and changes between censuses only across places. The same crosswalks give the Kreise of 1882 of Donges et al. (2023).

The estimates use the population of 1871 from GPOP, which covers the territory of today's Germany. Outside it, mostly in the east, the population is recorded as zero, so 135 of the 493 places of the design enter the estimates without a population trend; the hazard model marks these places with an indicator (Table C1). The population of 1871 in Figure 3 covers every place. For Prussia, it is the census of 1871 from the ifo Prussian Economic History Database; elsewhere, it is the population counted by the occupational census of 1882, projected back to 1871 with the growth of the place between the censuses of 1882 and 1895, capped at -1 and 3 percent a year. On the Prussian places, this projection misses the census of 1871 by a median of 3.7 log points. The census districts are mapped to the statistical units of 1900 by the overlap of their areas, weighted by population density.

A.3. Industrial and Traditional Patent Classes

Chemistry and electrical engineering are the largest classes of high-value patents; machine building, the Society's core field, is spread over several classes. Table A2 lists the 50 industrial and the 38 traditional classes into which the 89 classes of the Imperial Patent Office are grouped by their titles. Agriculture (45) is one exception: its title names a traditional field, but most of its high-value patents were farm machinery, so it counts as industrial. Bakery (2) and clothing (3) also count as industrial in this split; Table E3 moves them to the crafts. Class 62, salt works, holds 9 high-value patents and is counted with the traditional classes, as the note of the table explains.

A.4. Patent Texts: Coverage, Parameters, and Validation

The 64-dimensional embeddings come from the Google Patents Public Datasets (Abi Younes and de Rassenfosse, 2024).[19] They cover 290,385 of the 306,070 patents in the register (Griesshaber and Streb, 2026), 94.9 percent, and 99.9 percent of the patents located in a statistical unit. They order pairs of patents as expected (Table A3): Diesel's patent 82168 of 1895 has a similarity of 81 with his patent 67207, to which its claim refers, against 48 for random pairs, and close pairs are twice as frequent within a place as between territories: for a high-value patent, 2.0 percent of the patents granted in the five preceding years in its own

[19] Data set `google_patents_research`, table `publications`, field `embedding_v1`, queried with Google BigQuery in September 2026.

Table A2. The 89 Patent Classes by Group

Industrial classes (50)	Traditional classes (38)
1 Ore and fuel processing; 2 Bakery; 3 Clothing and apparel; 5 Mining; 7 Sheet metal, metal pipes and wire; 8 Bleaching, dyeing and textile finishing; 12 Chemical processes and apparatus; 13 Steam boilers; 14 Steam engines, including locomotive and marine engines; 15 Printing, typewriters and stamps; 17 Refrigeration, heat exchange and liquefaction; 18 Iron production; 19 Railway, road and bridge construction; 20 Railway operations; 21 Electrical engineering; 22 Paints, varnishes and adhesives; 23 Fat and oil industry; 24 Furnaces and firing systems; 26 Gas production; 27 Blowers and ventilation machines; 31 Foundry; 35 Hoisting equipment; 36 Heating and ventilation of buildings; 37 Building construction; 40 Metallurgy and alloys; 42 Instruments; 43 Control devices and coin-operated machines; 45 Agriculture, forestry, horticulture, viticulture and dairy; 46 Air and gas engines, spring and weight motors; 47 Machine elements; 48 Chemical metal processing; 49 Mechanical metal working; 52 Sewing and embroidery; 55 Paper and cardboard manufacturing; 58 Presses; 59 Pumps; 60 Governors for prime movers; 63 Carriages, motor vehicles and bicycles; 65 Shipbuilding and navigation; 67 Grinding and polishing; 73 Rope making; 74 Signaling; 76 Spinning; 80 Ceramics, stone, cement and briquetting; 82 Drying, kilning and centrifuges; 84 Hydraulic and foundation engineering; 85 Water supply and sewerage; 86 Weaving; 87 Tools, including pneumatic tools; 89 Sugar and starch production	4 Lighting and burners; 6 Beer, spirits, wine, vinegar and yeast; 9 Brushes and brooms; 10 Fuels; 11 Bookbinding; 16 Fertilizer; 25 Braiding, lacemaking and knitting; 28 Tanning and leather; 29 Textile fibers; 30 Health care; 32 Glass; 33 Hand and travel implements; 34 Household appliances and utensils; 38 Woodworking; 39 Horn, ivory, carving materials and rubber; 41 Hats and felt; 44 Haberdashery and smoking goods; 50 Milling and crushing machinery; 51 Musical instruments; 53 Food and luxury goods; 54 Paper and cardboard products; 56 Harnesses; 57 Photography; 61 Rescue and life-saving equipment; 64 Dispensing apparatus; 66 Slaughtering and meat processing; 68 Locksmith products and safes; 69 Cutting tools and edged weapons; 70 Writing and drawing instruments; 71 Footwear; 72 Firearms and fortification; 75 Sculpture, painting and surface decoration; 77 Sports, games, toys and aeronautics; 78 Explosives and matches; 79 Tobacco; 81 Transport and packaging; 83 Clocks and watches; 88 Wind and water power machines

Notes: The 89 classes of the German patent classification, with English short names of the class titles. Class 62 (Salinenwesen, salt works, in the classification of 1877) was discontinued in a revision of the classification, and the class list of the Patent Office used here marks it as missing; its 9 high-value patents, of 1878–1909, are counted with the traditional classes. Agriculture (45) is counted as industrial: of its 463 high-value patents in the places of the design that carry a subclass, 84 percent are for soil cultivation, sowing and fertilizing, harvesting, processing the harvest, or dairy machinery (subclasses 45a–c, 45e and 45g); the rest are for animal husbandry and fishing, traps and pest control, chemical agents, horticulture and forestry, and farriery.

place are close to it, against 1.2 percent of those of the other places of its territory and 1.0 percent of those of other territories.

For a patent granted in year t in a German place, I average its similarity to the patents granted in $t-5$ to $t-1$ within five groups: the patents of the same place, of the other places of its territory, of other territories, of other territories and the same German class, and of foreign or unlocated holders. The local premium is the first of these means minus the fourth. A pair is close if its similarity is at least 0.73 (73 after multiplying by 100), the 99th

Table A3. Similarity of Patent Pairs by Type of Pair

Type of pair	Pairs	Mean	Median	Share close
Random pairs	50,000	48.4	48.9	0.009
Same German class	50,000	58.7	59.5	0.101
Same subclass of the modern classification	50,000	62.3	63.5	0.176
Same holder	37,094	52.1	52.6	0.047
Patent and a patent named in its register entry	502	69.7	73.4	0.516
Addition patent and its parent patent	5,544	69.8	73.6	0.518
Known pairs: similarity (percentile among random pairs)				
Diesel 82168 (1895) and Diesel 67207 (1893), cited in the claim		81.4 (100)		
Diesel 82168 and Oechelhäuser 49935 (1889)		79.1 (99.9)		
Hoechst 236375 (1910) and 167410		79.8 (99.9)		
Benz 37435 (1886) and Diesel 82168		61.1 (85.9)		

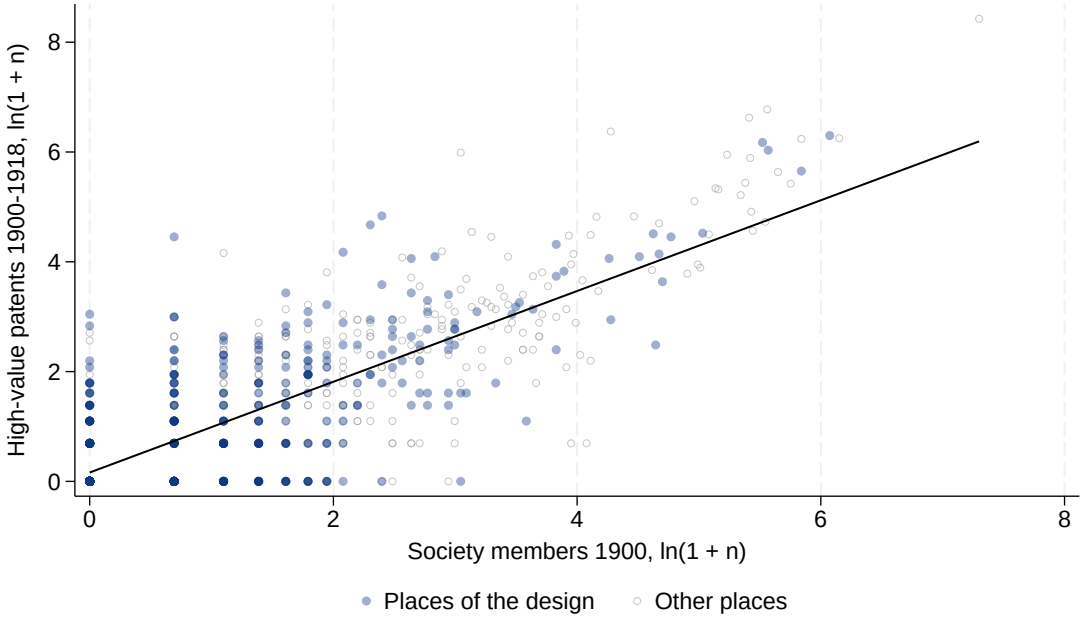
Notes: Cosine similarity ×100 of the Google Patents embeddings. Close: similarity of at least 73, the 99th percentile of random pairs. Same subclass: first CPC subclass of both patents. Patents named in a register entry and addition patents from the register (Griesshaber and Streb, 2026).

percentile of random pairs; all outcomes are multiplied by 100.

A.5. Society Members and High-Value Patents across Places

Places with more Society members in 1900 held more high-value patents in 1900–1918, within chapters (Figure 3) and across all places (Figure A1). Across the 493 places of the design, the slope is 0.793 (0.034) without controls and 0.649 (0.042) with chapter effects and the log population of 1871 from the censuses (Appendix A.2); per 10,000 inhabitants of 1871, in $\ln(1 + x)$ and without controls, it is 0.732 (0.046); with the GPOP population, across the 358 places where it is positive, 0.700 (0.058).

Figure A1. Society Members in 1900 and High-Value Patents of 1900–1918 across All Places



Notes: One dot per statistical unit of 1900 (1,020 units): ln(1 + high-value patents) of 1900–1918 against ln(1+Society members) of 1900, without controls. Filled dark blue: the 493 places of the design; hollow gray: all other places. 258 units with neither members nor high-value patents lie at the origin. Line: least-squares fit across all units, slope 0.826 (robust standard error 0.022); across the places of the design, the slope is 0.791 (0.034). Members: directory of 1900; high-value patents: Streb et al. (2006).

Appendix B. Estimators, the Comparison Group, and Robustness

Throughout, i indexes places (statistical units of 1900) and t years from 1877 to 1918; G_i is the founding year of the chapter whose territory covers place i, $D_{it} = \mathbf{1}[t \geq G_i]$, P_{it} the number of high-value patents held in place i and granted in year t, $Y_{it} = \ln(1 + P_{it})$, and Z_{it} the railway kilometers of place i in year t and the place's 1871 population times $(t - 1877)$. Standard errors are clustered by place unless stated otherwise.

B.1. Comparison Places by Cohort

Table B1 counts the not-yet-treated places by cohort (Section 5.2). In the largest cohort, that of 1892, the cohort-year effects lie between −0.02 and 0.04 in 1892–1903 and between 0.03 and 0.28 from 1904 on. In the register of all patents, the Mosel places are as flat over 1900–11 as in the high-value patents, while the other cohorts rise by 0.08 to 0.25.

Table B1. Places by Cohort: Opened, Treated by Then, and Not Yet Treated

		Places		
Cohort	Chapters	Opened	With a chapter by then	Not yet treated
1880	2	52	52	441
1882	2	46	98	395
1883	1	19	117	376
1888	1	21	138	355
1891	1	4	142	351
1892	2	113	255	238
1894	1	12	267	226
1897	1	9	276	217
1898	2	72	348	145
1899	1	8	356	137
1900	1	37	393	100
1902	1	17	410	83
1903	1	11	421	72
1904	3	49	470	23
1908	1	13	483	10
1912	1	10	493	0

Notes: The 493 places of the design (Table A1). Opened: places whose chapter opened in the year of the cohort. With a chapter by then: places whose chapter opened in that year or earlier. Not yet treated: places whose chapter opened later, the comparison group of the main design from the year of the cohort until the next cohort opens.

B.2. Designs with a Fixed Comparison Group

Designs with a fixed comparison group measure mainly the first decade after an opening, because the stacked design covers years 0 to 8 and the windows end before the late cohorts open. On the high-value patents, the stacked design with clean controls and $K = 8$ gives 0.015 (0.010), and 0.013 (0.007) without the chapter towns; the window to 1907, with the cohorts of 1908 and 1912 as controls, gives 0.009 (0.014) and the window to 1903, with the cohorts of 1904–12 as controls, −0.013 (0.016); on the window to 1903, the imputation estimator gives 0.011 (0.010) and a cohort × period regression 0.011 (0.009). Before founding, the stacked coefficients of years −5 to −2 average 0.002 (0.013), and the leads of the windows 0.001 (0.017) and 0.001 (0.016). With the 1882 structure held fixed, the window to 1907 gives −0.010 to 0.018. Over years 0 to 9, the main design gives 0.023 (0.015) and the window to 1907 0.012 (0.014), both on the 469 places of Table 2, Panel A. Estimated one cohort at a time, the stacked design gives effects between −0.04 and 0.16.

In the main design, the cohort-year cells up to 1903, which carry 60 percent of the weight of the estimate, average −0.015 (0.015); the cells of 1904–07, whose comparison group consists of the 23 places of the Bodensee and Mosel chapters, average 0.077 (0.021), and those of 1908–11, with the 10 places of the Mosel chapter, 0.152 (0.021). The rise in 1904 occurs in the same calendar year for cohorts at very different times since founding (year 24 for the cohort of 1880, year 2 for that of 1902) and coincides with the change in the comparison group. Without the Mosel cohort the estimate is 0.012 (0.016), without the Bodensee cohort 0.022 (0.016) and without any other single cohort between 0.032 and 0.047.

A window ends the sample in a year $\bar{T} \in \{1903, 1907\}$ before the last cohorts open; the cohorts with $G_i > \bar{T}$ are untreated throughout and form the comparison group. The stacked design builds, for each cohort g_s, a stack s of the places of that cohort and its clean controls, the places treated more than K years later and the 34 never-treated places outside the design, over the event years −5 to K:

$$Y_{ist} = \alpha_{is} + \lambda_{st} + \beta\, \mathbf{1}[G_i = g_s]\, \mathbf{1}[t \geq g_s] + \delta' Z_{it} + \varepsilon_{ist}, \qquad t \in [\, g_s - 5,\ g_s + K\,], \tag{B.1}$$

with stack-by-place and stack-by-year effects (Cengiz et al., 2019). The event-study version replaces the treatment term by event-time indicators; the main version uses $K = 8$.

The imputation estimator of Borusyak et al. (2024) fits $Y_{it} = \alpha_i + \lambda_t + \delta' Z_{it} + \varepsilon_{it}$ on the not-yet-treated place-years, imputes the untreated outcome of the treated place-years and averages the differences. The extended two-way fixed effects regression of Wooldridge (2025) gives every cohort-year after founding its own coefficient.

Figure B1 draws the event study of the imputation estimator. Its coefficients before

Table B2. Estimates by Comparison Group and Estimator

	(1) High-value patents	(2) All register patents	(3) Before founding
Panel A. Main design, places not yet treated as the comparison group			
All post-founding years	0.038***	0.025	0.006
	(0.014)	(0.035)	(0.018)
Adding the 34 never-treated places	0.013	0.005	0.006
	(0.018)	(0.029)	(0.017)
Never-treated places only	0.005	0.010	−0.020
	(0.024)	(0.035)	(0.028)
Panel B. Other estimators, all years			
Two-way fixed effects	0.015**	0.022	
	(0.007)	(0.017)	
Imputation	0.057***	0.034	0.021**
(Borusyak et al., 2024)	(0.011)	(0.029)	(0.009)

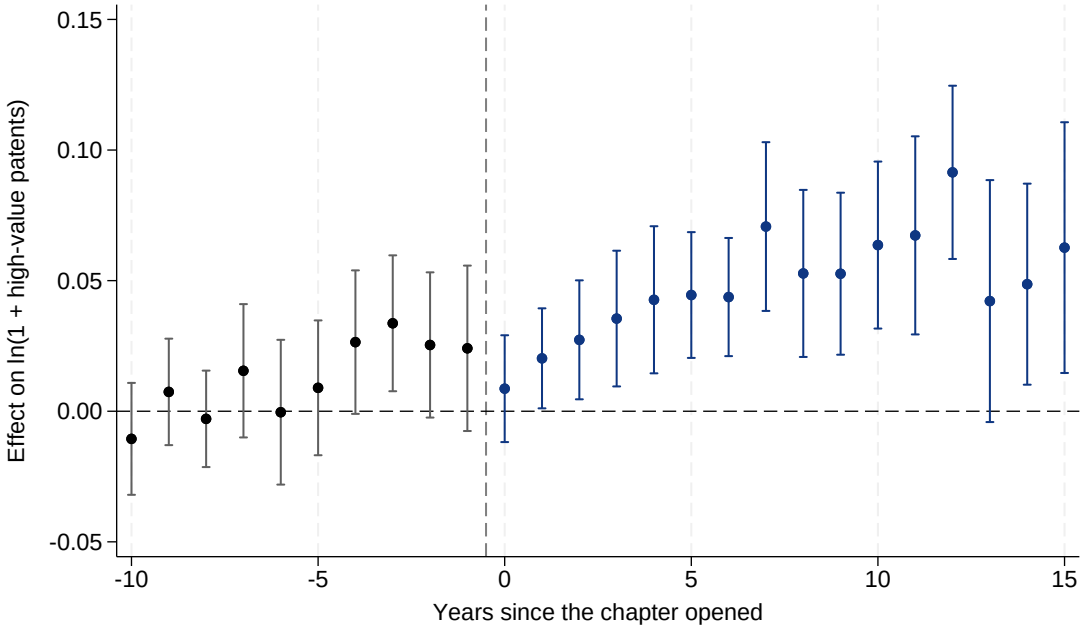
Notes: Column (1): ln(1 + high-value patents); column (2): ln(1 + all patents in the register) (Griesshaber and Streb, 2026), each patent located by the town of its holder as the high-value patents are, estimated on 1878–1916; column (3): average of the coefficients before founding, relative to the year before founding, over years −10 to −2 (estimator of Callaway and Sant'Anna (2021)), and for the imputation estimator the mean of its five coefficients before founding; Figure B1 draws the same regression with ten coefficients before founding. 493 places, 1877–1918, with railway kilometers and the 1871 population × trend. Panel A: doubly robust estimator of Callaway and Sant'Anna (2021); never-treated places: the 34 places outside the design whose territory had no chapter by 1918. Panel B: two-way fixed effects with place and year effects, and the imputation estimator on all years. Joint p-value of the coefficients of the main design before founding, years −16 to −2: 0.42. Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

founding come from a separate regression of the not-yet-treated place-years on indicators for the ten years before founding, relative to the place-years further from founding, the test the authors propose.

The estimator of de Chaisemartin and D'Haultfœuille (2026) is described with Figure 5, panel (a). The interaction-weighted estimator of Sun and Abraham (2021) uses the Mosel cohort of 1912 as the control cohort on the years 1877–1911, so that the control cohort is untreated throughout. Over years 0 to 15, the coefficients average 0.048 (0.008) with the imputation estimator and 0.044 (0.016) with that of Sun and Abraham (2021); the estimator of de Chaisemartin and D'Haultfœuille (2026) gives an average effect of 0.045 (0.013). Before founding, its 14 placebos average −0.002 and are jointly insignificant ($p = 0.47$), and the Sun–Abraham coefficients average −0.017 (0.018). The ten imputation leads average 0.013 (0.008); they lie close to zero from year −10 to −5 and at 0.02 to 0.03 in the four years before founding, when, in the hazard model, openings followed a local high in patenting

(Appendix C.2).

Figure B1. Event Study with the Imputation Estimator



Notes: Coefficients on ln(1+high-value patents) by years since the chapter opened, with 95 percent confidence intervals; 493 places, railway kilometers and the 1871 population × trend; standard errors clustered by place. Imputation estimator of Borusyak et al. (2024): place and year effects and the covariates fitted on the not-yet-treated place-years; from founding on, the average difference between the actual and the imputed outcome; before founding, the coefficients of a regression of the not-yet-treated place-years on indicators for the ten years before founding, relative to the place-years more than ten years before founding.

B.3. Count Models and Synthetic Difference-in-Differences

In Poisson models of the count P_{it}, high-value patents rose by 22 percent after an opening with place and year effects and by 34 percent with cohort-by-period effects (Table B3, Panel A). The models, $E[P_{it} \mid \alpha_i, \lambda_t, Z_{it}] = \exp(\alpha_i + \lambda_t + \beta D_{it} + \delta' Z_{it})$, assume parallel trends in proportions and avoid the dependence of $\ln(1 + P)$ on the units of a count with many zeros (Chen and Roth, 2024).

Synthetic difference-in-differences (Arkhangelsky et al., 2021), which weights the 34 never-treated places so that their path before founding parallels that of each cohort and weights the years before founding to resemble the years after, gives 0.030 to 0.038 (Panel B). With a linear trend for each place, the imputation estimator is uninformative, −0.024 (0.054).

Table B3. Poisson and Synthetic Difference-in-Differences Estimates

	Estimate	Effect in percent
Panel A. Poisson, count of high-value patents, 216 places		
Two-way fixed effects, all places	0.203^{***}	22
	(0.076)	
Cohort × period effects, not-yet-treated places	0.296^{**}	34
	(0.146)	
window to 1903	0.289^{*}	34
	(0.164)	
Panel B. Synthetic difference-in-differences, $\ln(1 +$ *high-value patents*$)$		
No covariates	0.030	
	(0.019)	
Railways and population × trend projected out	0.038^{*}	
	(0.020)	
Cohorts with at least ten years before founding	0.038^{**}	
	(0.015)	

Notes: Panel A: Poisson pseudo-maximum likelihood of the count of high-value patents on the 216 places with a high-value patent in 1877–1918, with place and year effects or cohort × period effects (Wooldridge, 2023); effect in percent: $\exp(b) - 1$. Panel B: synthetic difference-in-differences (Arkhangelsky et al., 2021) with the 34 never-treated places, reweighted, as the comparison group; 200 bootstrap draws. Railway kilometers and the 1871 population × trend. Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B.4. Other Samples and Scales

The estimate stays positive and significant at the 5 percent level in every other sample and scale (Table B4, Panel A). Without the five eastern chapters it is 0.048 (0.018), and in the balanced panel of the cohorts observed for at least ten years before founding 0.065 (0.016); ending the sample in 1913 leaves it unchanged, because no place is untreated after 1911 and the main design uses no later year (Panel B).

Table B4. Other Samples and Scales

Specification	Estimate (SE)	Note
Panel A. Other samples and scales		
All 493 places, 1877–1918	0.038^{***} (0.014)	
Inverse hyperbolic sine of high-value patents	0.047^{***} (0.018)	
469 places with the population density of 1900	0.040^{***} (0.014)	
Kreise of 1882 as units	0.074^{***} (0.021)	365 Kreise
Alternative founding dates	0.043^{***} (0.015)	chapters with breaks (Appendix A.1)
Years 0–15 only	0.030^{**} (0.014)	
Panel B. Further samples		
Before the war, 1877–1913	0.038^{***} (0.014)	the main estimate: no place is untreated after 1911
Without the eastern territories	0.048^{***} (0.018)	13 cohorts, 362 places
Balanced panel, at least 10 years before founding	0.065^{***} (0.016)	cohorts of 1888–1912, 376 places

Notes: Estimator of Callaway and Sant'Anna (2021) with the not-yet-treated places as the comparison group; railway kilometers and 1871 population × trend; standard errors clustered by place in parentheses. Panel B: before the war: years 1877–1913; eastern territories: the chapters of East Prussia, West Prussia, Posen, the Mark, and the Lausitz, whose places are dropped, which leaves 13 cohorts; balanced panel: the cohorts of 1888 and later, each observed for at least ten years before founding. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix C. Pre-Trends and Selection into Founding

C.1. Pre-Trends: Residual Drift and Bounds on Violations of Parallel Trends

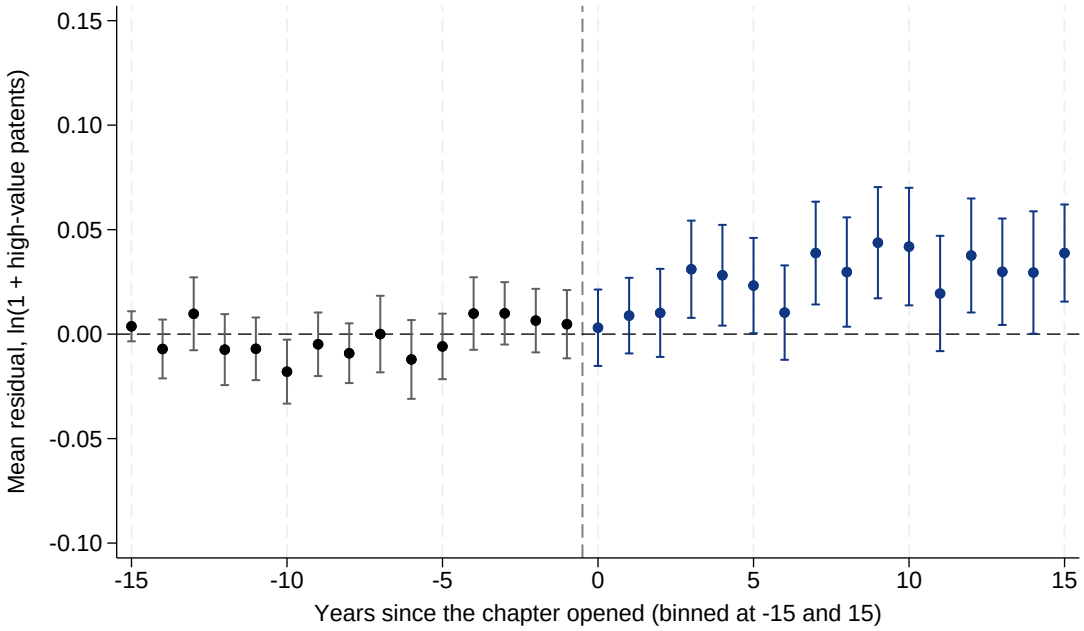
Figure C1 follows the logic of the imputation estimator with the 1882 structure held fixed: place effects, year effects, railway kilometers, the 1871 population × trend, and the 1882 industry share, agriculture share, and machine workers × trend are estimated on the not-yet-treated place-years, and the residuals of the chapter places are followed by years since founding. Before founding the residuals average zero by construction, so their slope is the test; after founding they measure the gap between the chapter places and their imputed path. Net of these effects and controls, the chapter places drift upward before founding by 0.0010 a year (standard error 0.0006), and the ten leads are jointly insignificant ($p = 0.17$); with railway kilometers and the population trend alone, the drift is 0.0011 (0.0006) and $p = 0.15$. Extrapolated, the drift would account for about 0.005 to 0.006 of the average effect over years 0 to 9. After founding the residuals average 0.027, rising from about year 3; with standard errors clustered by founding cohort, the chapter places lie 0.041 (0.012) above the comparison places after founding and 0.030 (0.015) above the extrapolated drift ($p = 0.06$). In raw means (Figure 4), the gap between the chapter places and the not-yet-treated places rises by 0.0017 (0.0010) a year before founding and averages 0.058 (0.031) after.

Under the relative magnitudes restriction of Rambachan and Roth (2023), the average effect over years 6 to 12 excludes zero up to $\bar{M} = 0.05$, with a 95 percent interval from 0.003 to 0.087; $\bar{M}$ bounds the change in the violation of parallel trends between consecutive years after founding relative to the largest such change before founding. The average over years 0 to 12 excludes zero only at $\bar{M} = 0$, and under the smoothness restriction the intervals include zero already at $M = 0$, because the coefficients before founding are imprecise.

C.2. Selection into Founding: Level, Growth, and Distance

In the discrete-time hazard model (Table C1), every place that is not yet a chapter town is at risk in each year from 1883 to 1914. The level of patenting over the previous three years predicts an opening in every column and the distance to the nearest chapter in every linear column, while growth over three, five, and ten years enters negatively; the 1882 industry share adds to the level. Railway kilometers, held fixed in every column, enter negatively, −0.38 (0.22) in column (1) and significantly so when growth is measured over five or ten years, −0.55 (0.19) and −0.64 (0.23): given the level of patenting and the population, places on more railway lines did not open a chapter earlier.

Figure C1. Residuals of the Chapter Places around Founding, 1882 Structure Held Fixed



Notes: Mean residuals of ln(1 + high-value patents) of the chapter places by years since their chapter opened, with 95 percent confidence intervals, standard errors clustered by place. Residuals from place effects, year effects and the covariates of Appendix C.1, all fitted by least squares on the not-yet-treated place-years only; places of the design with 1882 census data; years before −15 and after 15 binned. Because the place effects are fitted on each chapter place's own years before founding, its residuals before founding average zero by construction, so the test is their slope. With railway kilometers and the population trend alone, the path after founding is that of Figure B1.

Early and late chapter places differed little in 1882 (Table C2): the founding year does not follow the industry share, and a standard deviation more machine workers goes with a chapter about one year earlier. The 1882 structure predicts a gap in patent trends between early and late places of 0.0003 a year in absolute value, about 0.002 when extrapolated over years 0 to 9 as in Appendix C.1, a twentieth of the estimate on all 493 places.

For the twelve chapters opened after the census of 1895, the change of the engineering industries and of the industry share between 1882 and 1895 did not predict whether or when a chapter opened. The places whose chapter opened after 1895 did not differ from the places without a chapter by 1918 in either change, −0.08 (0.12) and −0.009 (0.008), with the start level, population, and state held fixed (wild-cluster bootstrap p-values 0.64 and 0.65). The founding year does not follow these changes ($p = 0.14$ and 0.76), and their growth did not change when a chapter opened ($p = 0.58$ and 0.32). Against the two latest chapters alone, the industry shares of the cohorts of 1897–1904 grew 0.009 (0.004) faster, with a bootstrap p-value of 0.57.

Table C1. Hazard of Becoming a Chapter Town, 1883–1914

	Linear probability, openings per 1,000 places				Logit
	(1)	(2)	(3)	(4)	(5)
Level of patenting, previous three years	1.72**	1.42**	1.60**	1.55**	0.66***
	(0.67)	(0.56)	(0.69)	(0.66)	(0.24)
Growth of patenting, three years before	−0.87**			−0.82**	−0.32
	(0.42)			(0.42)	(0.22)
Growth of patenting, five years before		−0.71*			
		(0.40)			
Growth of patenting, ten years before			−0.78		
			(0.60)		
Distance to the nearest existing chapter (log)	0.47**	0.47**	0.55**	0.66***	0.76*
	(0.19)	(0.19)	(0.24)	(0.20)	(0.46)
Population 1871 (log)	1.75	5.20***	6.39***	1.33	1.81
	(3.48)	(1.92)	(2.34)	(3.45)	(4.71)
Industry share, 1882				0.45**	
				(0.21)	
Machine workers per 1,000, 1882				0.24	
				(0.20)	
Place-years	32,147	30,122	25,063	32,147	32,147

Notes: Discrete-time hazard of a place becoming a chapter town, 1883–1914; 1,013 places, 17 openings (the chapters of 1883–1912 other than Posen, which is not on the 1925 map), base rate 0.53 openings per 1,000 places a year. Coefficients per standard deviation of each variable. Every column also holds fixed the place's railway kilometers of the previous year, which enter negatively (−0.30 to −0.64 per standard deviation in columns (1)–(4), significant at the 1 percent level in columns (2) and (3)), and an indicator for places without an 1871 population. Column (4): the 980 places with 1882 census data. Columns (1)–(4): linear probability models scaled to openings per 1,000 places a year, with year effects; column (3) on the years from 1890. Column (5): logit with five-year period effects (log odds). Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C2. Places of the Design in 1882, by Decade of Founding

	Chapter founded			Later minus earlier
Place in 1882	1880–89	1890–99	1900–12	per decade (SE)
Industry share of the workforce	0.29	0.34	0.28	0.014 (0.023)
Agriculture share of the workforce	0.57	0.52	0.56	−0.019 (0.023)
Machine and instrument workers per 1,000	6.8	5.8	4.8	−0.508 (0.778)
Population (log)	10.61	10.26	10.33	−0.131 (0.086)
ln(1 + high-value patents), 1877–79	0.11	0.11	0.10	0.005 (0.021)
Founding years	4	6	6	
Places	133	208	128	

Notes: Means in 1882 of the places of the design, by the decade in which their chapter opened; the 469 places of Table 2, Panel A. High-value patents: average of 1877–79. Last column: each characteristic regressed on the founding year, scaled to one decade, with standard errors clustered by founding cohort. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C.3. Chapter Towns and Their Territories

The gain of the chapter town on its territory declines with distance and had begun before founding. The comparison of the main text regresses $\ln(1 + P_{it})$ for the places of the 22 territories on place effects, territory-by-year effects, the controls, and indicators for the chapter town and for places 0 to 15 and 15 to 40 km from it after founding, with the places more than 40 km away as the reference. The chapter town gains 0.381 (0.065), places 0 to 15 km away 0.045 (0.037) and places 15 to 40 km away 0.029 (0.016); with the population density of 1900 $\times$ trend added, the gains are 0.323 (0.059), 0.039 (0.040) and 0.027 (0.017), and the chapter town's gain is 0.323 (0.057) with standard errors clustered by territory.

In the event study of the chapter town against the places more than 40 km away, the nine coefficients before founding are jointly significant ($F = 5.16$, $p < 0.001$), and the coefficients rise by about 0.041 a year over the nine years before founding and by about 0.014 a year over the fifteen years after it. Across all places, two-way fixed effects give a gain that declines from 0.377 (0.064) in the chapter town to 0.059 within 15 km and about zero beyond 40 km.

Appendix D. Prior Industrialization, the Inflow of Engineers, Rings, and Spillovers

D.1. Prior Industrialization Held Fixed: The Doubly Robust Estimator

Whereas the imputation estimator is stable across the sets of 1882 controls (Table 3), the doubly robust estimator with the full 1882 set has too few comparison places per cell to inform: with railways, population, density, the 1882 shares, machine workers, and population 1882, it gives 0.004 (0.037) on the statistical units and 0.001 (0.066) on the Kreise, against 0.030 (0.015) and 0.007 (0.039) with the shares alone, and the outcome-regression version of the estimator gives 0.033 (0.038) and −0.019 (0.071) with the full set and 0.061 (0.016) and 0.013 (0.028) with the shares. With the full set, the doubly robust estimator's cells up to 1903 average about −0.011, close to the main specification, while the cell of 1911, with the 10 Mosel places as the only comparison places, is −0.030 (0.639); with the sample ending in 1907, the estimate is −0.010 (0.017).

With the variables of column (5) of Panel C of Table 2, the Society members and the members in no chapter of 1900 and the 1882 industry share, machine workers, and population, the doubly robust estimator gives 0.020 (0.077) on the seven chapters opened after the directory of 1900, against 0.084 (0.031) from the imputation estimator. Standardizing the 1882 variables within each of the two census sources leaves the estimates within their standard errors.

D.2. The Stock of Society Members and Founding Rolls

The seven chapters founded after the directory of 1900 took over organized and unorganized members alike: of the 394 Society members in their territories in 1900, 28 percent belonged to no chapter, against 11 percent elsewhere, and 62 percent were registered with the nearest existing chapter, on average 54 km away. Some of the seven, such as the Lausitz, organized regions without a chapter; others, such as Augsburg and the Emscher, were carved out of a large neighboring territory. Their founding year does not follow the member density of 1900 (0.21 years per unit of log density, standard error 0.23).

Later chapters started with larger founding rolls in smaller towns: across the 44 chapters opened between 1856 and 1914 that the yearbook of 1925 lists, the median founding roll was 23 members before 1880, 87 in 1880–99, and 128 after 1900, its share of the Society's membership fell from 3.4 to 0.6 percent, and the median chapter town had 156, 101, and 31 Society members in 1900. Given the founding year, the size of the chapter town does

not predict the founding roll (+0.09 per log member, standard error 0.11). Neither pattern suggests that a chapter opened once a region reached a threshold number of engineers.

D.3. Inflow of Engineers and Industry, and Other Changes of the Period

No stock of engineers or industry grew significantly faster where a chapter opened, in regressions of the change of a stock over a directory or census window on an indicator for a chapter opening inside the window (Table D1). The agriculture share fell faster between 1882 and 1895 where a chapter opened (−0.016, standard error 0.004), but not once the start level, population, and state are held fixed (−0.003, 0.009), and between 1895 and 1907 it fell less where a chapter opened, the opposite of an inflow of industry.

Table D1. Growth of Engineers and Industry Where a Chapter Opened

Stock	Window	(1) Chapter opened	(2) With start level, population, state
Society members per 1,000 inhabitants (log)	1900–1925	0.04	0.06
		(0.08)	(0.08)
Society members in no chapter (log)	1900–1925	0.02	−0.01
		(0.04)	(0.04)
Engineering-industry workers (log)	1895–1907	−0.03	−0.09
		(0.05)	(0.07) [0.34]
Industry share of the workforce	1895–1907	0.0008	−0.0027
		(0.0028)	(0.0032) [0.43]
Agriculture share of the workforce	1895–1907	−0.0005	0.0080**
		(0.0042)	(0.0040) [0.07]
Engineering-industry workers (log)	1882–1895	−0.03	0.01
		(0.03)	(0.03)
Industry share of the workforce	1882–1895	0.0042	−0.0014
		(0.0034)	(0.0066)
Agriculture share of the workforce	1882–1895	−0.0159***	−0.0033
		(0.0037)	(0.0087)

Notes: Change of the stock over the window, $\ln S_{it_1} - \ln S_{it_0}$ (shares in levels), regressed on an indicator for a chapter opening in the window, against every other place. Column (2) holds the start level, the population, and state effects fixed. Society-member rows: 468 places in the territories of 22 chapters (469 for members in no chapter in column (2)); column (1) holds the member density of 1900 and the population of 1882 fixed, column (2) the start level, the population of 1882, and state effects; standard errors clustered by chapter. Engineering industries: machine, instrument, and electrical industries. The census of 1907 counts family helpers more completely than earlier censuses. Robust standard errors in parentheses, wild-cluster bootstrap p-values in brackets where shown; stars from the robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Other changes of the period could have raised high-value patents where chapters opened. A chapter was also a showroom for members' designs, and a patent known to potential buyers was worth renewing, so part of the rise may reflect the market for patents, in which about

8 percent of patents changed owner at least once (Burhop, 2010). Invention in chemicals and electrical engineering moved into firm laboratories (Meyer-Thurow, 1982; Burhop and Lübbers, 2010); a laboratory in a chapter town would raise that town's patents, which the estimates without chapter towns address. Branches of the chemists' and electrical engineers' societies that opened together with a chapter are part of the estimate, while cartels and trade associations, organized by industry across regions, cannot produce a rise dated to a regional opening.

D.4. Rings around Chapter Towns and Schools

The ring designs compare places 10 to 50 km from a town with places 50 to 100 km away, with the town itself dropped: around the chapter towns, with the founding of the chapter as the event, and around the 24 state machine-building schools with a dated opening, with the opening of the school as the event. Each statistical unit of 1900 is assigned to its nearest chapter town or school; units whose nearest town or school opened before 1878 drop out of the inner ring, and the outer ring is never treated. Unlike the main design, the ring designs therefore have a comparison group that is never treated, which lets the imputation estimator fit the year effects on all years up to 1918.

Around chapter towns, the imputation estimator gives 0.021 (0.011) over all years after the opening, 0.008 (0.009) in years 0 to 7 and 0.047 (0.015) in years 8 to 15 (Table D2 and Figure D1); the doubly robust estimator of Callaway and Sant'Anna (2021) gives 0.012 (0.019) over all years. Around the schools, the two estimators give 0.028 (0.019) and 0.014 (0.026); the imputation estimates are 0.023 (0.013) in years 0 to 7 and 0.018 (0.023) in years 8 to 15, with no rise in later years. Before the opening, the coefficients of both designs lie close to zero: with the imputation estimator, the ten years before the opening average 0.001 (0.008) around chapter towns and −0.003 (0.012) around schools (joint tests: $p = 0.83$ and $p = 0.42$).

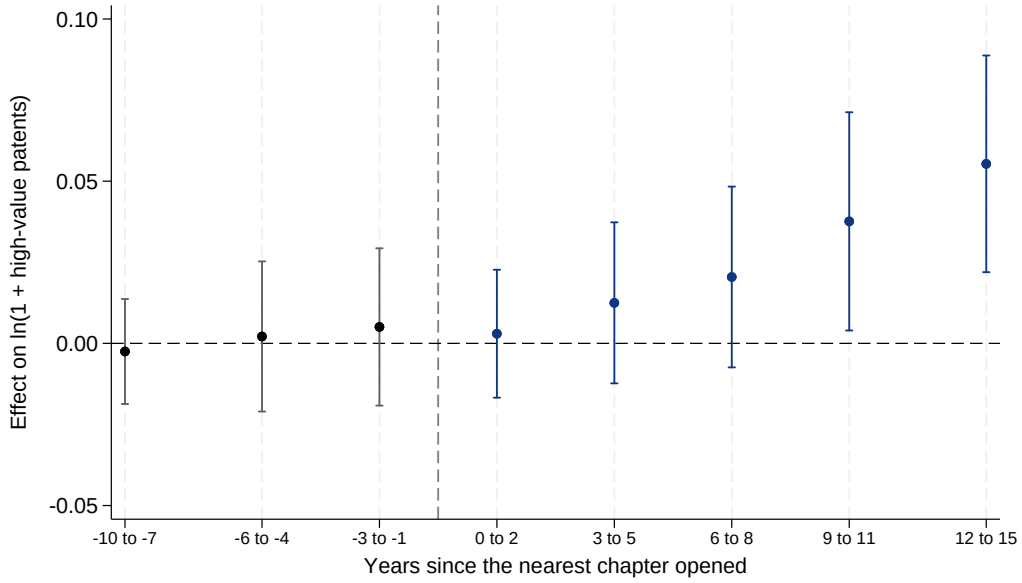
The rings are a weak test by construction: the outer ring of one chapter town is often the inner ring of the next, the places 10 to 50 km from a chapter town held few Society members (most members lived in the chapter town itself, which the design drops), and the territories of the main design extend beyond 50 km from their town, so that both rings are treated in the main design.

D.5. Spillovers and the No-Interference Assumption

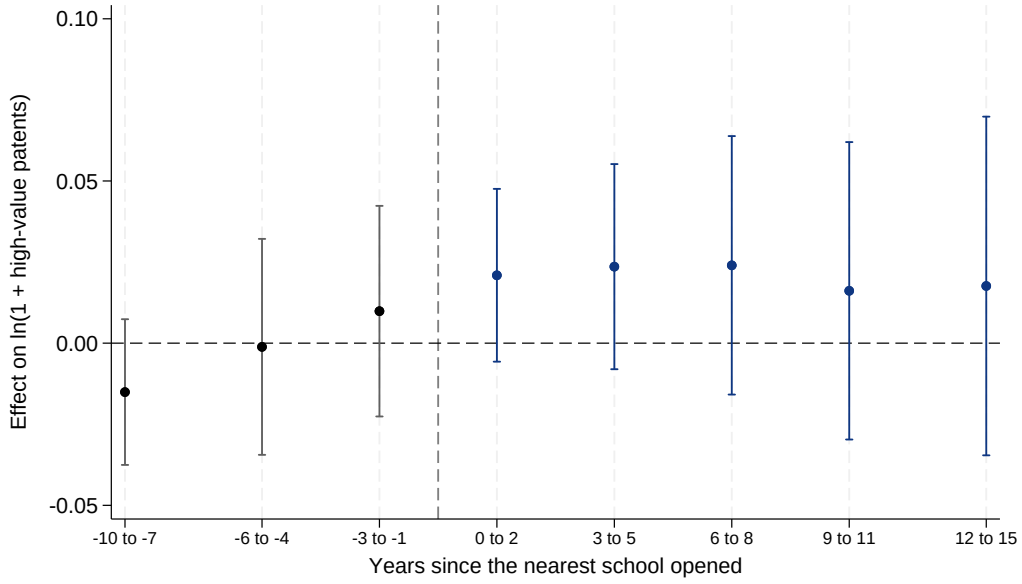
The design identifies the effect of an opening only if the opening left the patenting of the comparison places unchanged: the outcome of a place may depend on whether its own

Figure D1. Event Studies of the Rings around Chapter Towns and Machine-Building Schools

(a) Rings around chapter towns



(b) Rings around machine-building schools



Notes: Coefficients on ln(1 + high-value patents) of the places 10–50 km from the nearest chapter town (panel a) or state machine-building school (panel b), relative to the places 50–100 km away, by years since the opening, in bins of three years (the first and the last bin four), with 95 percent confidence intervals; estimator, covariates, and pre-opening coefficients as in Panel A of Table D2; from the opening on in dark blue; the same vertical axis in both panels; standard errors clustered by place.

Table D2. Ring Estimates around Chapter Towns and Machine-Building Schools

	Dependent variable: ln(1 + high-value patents)	
	Rings around chapter towns	Rings around machine-building schools
Panel A. Imputation estimator		
All years after the opening	0.022^{**}	0.028
	(0.011)	(0.019)
Years 0–7	0.008	0.023^{*}
	(0.009)	(0.013)
Years 8–15	0.047^{***}	0.018
	(0.015)	(0.023)
Before the opening, years −10 to −1	0.001	−0.003
	(0.008)	(0.012)
Joint test of the years before the opening, *p*-value	0.83	0.42
Panel B. Estimator of Callaway and Sant'Anna		
All years after the opening	0.011	0.014
	(0.019)	(0.026)
Before the opening, year-on-year changes	0.002	0.002
	(0.001)	(0.003)
Places in the inner ring, 10–50 km	227	213
Places in the outer ring, 50–100 km	364	381
Opening years, 1878–1918	16	11

Notes: Places 10–50 km (inner ring) and 50–100 km (outer ring) from the nearest chapter town or state machine-building school, the town itself (within 10 km) dropped; all statistical units of 1900, 1877–1918. Event: the founding of the nearest chapter or the opening of the nearest school in 1878–1918; units nearest to a town or school that opened before 1878 drop out of the inner ring; the outer ring is never treated. Railway kilometers and the 1871 population × trend. Panel A: imputation estimator of Borusyak et al. (2024), with place and year effects and the covariates fitted on the untreated place-years (the outer ring throughout, the inner ring before the opening); all years after the opening: the average over all treated place-years; years 0–7 and 8–15: averages of the event-study coefficients; before the opening: the average of the coefficients of a regression of the untreated place-years on indicators for the ten years before the opening, relative to the untreated place-years further back, and the joint test of the ten coefficients. Panel B: estimator of Callaway and Sant'Anna (2021) with the not-yet-treated places and the outer ring as the comparison group: simple aggregation of the cohort-year effects after the opening; before the opening: the average of the year-on-year coefficients. Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

chapter has opened, but not on whether the chapters of other places have. Three features of the setting bear on this assumption.

First, treatment is assigned to territories, not to single places. An opening reached all places of its territory at once, so that interactions among them, such as a chapter town that drew inventive work from its surroundings, are part of the estimated effect of an opening on

its territory rather than a violation of the assumption. Within territories, the places other than the chapter town gained as well (Appendix C.3), and the results do not rest on the chapter towns (Section 6.3).

Second, the comparison places were not untouched by the Society. Before membership in a chapter became compulsory in 1911, members could belong to a distant chapter or to none, and before a chapter opened, the members of its future territory were registered with chapters elsewhere, most often the nearest existing one (Appendix D.2). If the not-yet-treated places drew on the meetings of a neighboring chapter, or if their engineers took part in a chapter that opened nearby, the comparison places were partly treated, and the estimates understate the effect of an opening.

Third, an opening could have drawn inventors and firms, or the registration of their patents, into the new territory, at the expense of the not-yet-treated places. This would overstate the effect. Society members, engineering workers and the industry share did not grow significantly faster in the territories that received a chapter, but the intervals are wide (Table D1).

The direct tests, on the design of the main text, point to a bias in the direction of the third feature. Removing every place within 25 km of a place whose chapter had opened earlier, so that no such place serves as a comparison after a neighbor's chapter has opened, leaves the estimate on all places, 0.038, at 0.037 (0.015; 460 places); removing every place within 50 km lowers it to 0.026 (0.016; 401 places in 15 cohorts), part of which may reflect the smaller sample. In two-way fixed effects regressions on the 493 places, with place and year effects, railway kilometers, and the 1871 population $\times$ trend, not-yet-treated places whose nearest opened place lies 0 to 25, 25 to 50, or 50 to 100 km away have $\ln(1 + \text{high-value patents})$ lower by 0.029 (0.021), 0.024 (0.017), and 0.021 (0.011) than places with no opened place within 100 km ($p = 0.17$, 0.15, and 0.05; with standard errors clustered by territory, 0.24, 0.05 and 0.05). With these bands, the coefficient of an own opening is 0.001 (0.009), against 0.015 (0.007) without them (Table B2). Because the gap hardly changes with distance, it fits a slower rise of the late cohorts as well as displacement by the new territories. The share of neighbors 10 to 50 km away that lie in an opened territory has a coefficient of -0.031 (0.031) among the not-yet-treated places and 0.010 (0.026) among all 549 places outside an opened territory. None of these tests can detect competition among firms in national product markets, in which the firms of a new territory could have gained at the expense of firms elsewhere, so the estimates are effects relative to the not-yet-treated places.

Appendix E. What Changed: Heterogeneity and Patent Texts

E.1. Heterogeneity by Industrial Structure and Patent Class

The rise was larger in the more industrial thirds of the 1882 industry share (Table E1; each third is estimated against its own not-yet-treated places, which from 1904 on are the Bodensee and Mosel places that fall into the third). In the most industrial third, high-value patents rose by 0.166 (0.055) in years 6 to 12. Its coefficients before founding are negative and imprecise, −0.060 (0.041) on average, and drift upward by about 0.005 a year; extrapolated to years 6 to 12, this drift would reach a quarter to a third of the estimate for those years. In the middle third, the coefficients before founding average 0.039 (0.020). Two-way fixed effects with continuous interactions give the same ordering as the thirds: 0.067 (0.010) per standard deviation of the industry share, −0.069 (0.010) per standard deviation of the agriculture share and 0.038 (0.013) per standard deviation of machine workers.

Table E1. Estimates by Third of the 1882 Industry Share

	Statistical units of 1900			Kreise of 1882		
	Least	Middle	Most	Least	Middle	Most
All post-founding years	-0.026^{**}	0.055^{***}	0.105^{**}	0.002	−0.012	0.141^{***}
	(0.012)	(0.017)	(0.042)	(0.032)	(0.030)	(0.046)
Years 6–12	−0.009	0.067^{***}	0.166^{***}	0.057	0.040	0.191^{***}
	(0.011)	(0.021)	(0.055)	(0.038)	(0.037)	(0.057)
Before founding, years −11 to −2	−0.005	0.039^{*}	−0.061	0.021	0.002	−0.067
	(0.006)	(0.020)	(0.041)	(0.018)	(0.029)	(0.045)
Places	165	164	164	122	122	121

Notes: Estimator of Callaway and Sant'Anna (2021), doubly robust, on ln(1 + high-value patents) with railway kilometers and the 1871 population × trend, estimated separately within each third of the 1882 industry share, each third against its own not-yet-treated places; 1877–1918. Columns: the statistical units of 1900 with 1882 census data and the 365 Kreise of 1882 of Donges et al. (2023). Years 6–12 and before founding: averages of event-study coefficients relative to the year before founding. Standard errors clustered by place (Kreise: by Kreis) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Thirds of the 1882 agriculture share mirror the thirds of the industry share (Figure E1): in the least agricultural third, the estimate is 0.058 (0.045) on the statistical units and 0.134 (0.047) on the Kreise of 1882, in the most agricultural third 0.007 (0.009) and 0.027 (0.025). By thirds of machine workers per 1,000 inhabitants in 1882, the estimates range from 0.01 to 0.04 on the statistical units and from −0.06 to 0.15 on the Kreise.

By patent class (Tables E2 and E3), the rise is concentrated in the industrial classes. In

Table E2. Estimates by Group of Patent Classes

		Groups of Table A2		Crafts and all other		
	All classes	Industrial (50)	Traditional (38)	Crafts and consumer goods (25)	All other classes (64)	Active classes
	(1)	(2)	(3)	(4)	(5)	(6)
All post-founding years	0.038***	0.034***	0.012	0.007	0.036***	0.031**
	(0.014)	(0.012)	(0.010)	(0.009)	(0.013)	(0.013)
Years 6–12	0.042***	0.034***	0.015	0.012	0.036***	0.038***
	(0.015)	(0.013)	(0.011)	(0.009)	(0.014)	(0.015)
Before founding, years −11 to −2	0.007	0.018	−0.010	−0.003	0.011	0.004
	(0.018)	(0.015)	(0.010)	(0.010)	(0.015)	(0.016)
Places	493	493	493	493	493	493

Notes: Estimator of Callaway and Sant'Anna (2021), doubly robust, with railway kilometers and the 1871 population × trend and the places whose chapter has not yet opened as the comparison group; the 493 places of the design, 1877–1918. Outcomes: $\ln(1+x)$ of the high-value patents in all 89 classes, in the 50 industrial and in the 38 traditional classes (Table A2), in the 25 classes of crafts and consumer goods and in the other 64 classes, and of the number of classes with a high-value patent in the place and year (active classes). Crafts and consumer goods: the 25 classes of Panel C of Table E3. Years 6–12 and before founding: averages of the event-study coefficients of years 6 to 12 and −11 to −2, relative to the year before founding (Table B2 and Figure 5 average years −10 to −2). Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the split of Table A2, high-value patents rose by 0.034 (0.012) in the 50 industrial classes against 0.012 (0.010) in the traditional classes, which is not significant; that traditional group also holds fuels, glass, and milling. Table E3 redefines it in two steps from the class titles, so that bakery (2) and clothing (3), industrial in Table A2, count as crafts. The intermediate definition takes the 25 classes of crafts and consumer goods and adds fertilizer (16), agriculture, forestry, horticulture, viticulture and dairy (45), counted here by its title, and sugar and starch (89); fat and oil (23) and milling (50) count as industrial, because they also hold mineral oils and lubricants and general-purpose crushing machinery. The narrowest definition keeps the crafts and consumer goods alone. The traditional estimate is 0.012 (0.010), 0.013 (0.009) and 0.007 (0.009) under the three definitions, and the difference between the traditional and the industrial classes, estimated as its own outcome, is −0.022 (0.015), −0.020 (0.013) and −0.029 (0.015). With the industrialization of 1882 held fixed (imputation estimator, 469 places), the traditional classes rise by 0.018 (0.007) in the split of Table A2 and by 0.009 (0.007) in crafts and consumer goods, and the differences are −0.026 (0.013) and −0.041 (0.013). The number of classes with a high-value patent rose by 0.031 (0.013). Before founding, no coefficient differs significantly from zero.

In counts, the industrial classes account for 76 to 84 percent of the additional high-value patents under the three definitions, more than their 61 to 75 percent share before the openings, but the excess is not significant ($p = 0.13$, 0.61 and 0.24; Figure E2). Chemical

Table E3. Industrial and Traditional Classes under Three Definitions

	Industrial classes (1)	Traditional classes (2)	Traditional minus industrial (3)
Panel A. Split of Table A2: 50 industrial and 38 traditional classes			
All post-founding years	0.034^{***}	0.012	−0.022
	(0.012)	(0.010)	(0.015)
Years 6–12	0.034^{***}	0.015	−0.020
	(0.013)	(0.011)	(0.017)
Before founding (leads)	0.018	−0.010	−0.027
	(0.015)	(0.010)	(0.018)
High-value patents in the group	19,083	6,075	
Classes with high-value patents	50	39	
Panel B. Traditional: crafts, consumer goods, and agriculture (28 classes)			
All post-founding years	0.033^{***}	0.013	−0.020
	(0.012)	(0.009)	(0.013)
Years 6–12	0.034^{**}	0.017^{*}	−0.017
	(0.014)	(0.009)	(0.015)
Before founding (leads)	0.012	−0.003	−0.015
	(0.014)	(0.012)	(0.018)
High-value patents in the group	20,976	4,182	
Classes with high-value patents	61	28	
Panel C. Traditional: crafts and consumer goods (25 classes)			
All post-founding years	0.036^{***}	0.007	-0.029^{**}
	(0.013)	(0.009)	(0.015)
Years 6–12	0.036^{***}	0.012	−0.025
	(0.014)	(0.009)	(0.016)
Before founding (leads)	0.011	−0.003	−0.014
	(0.015)	(0.010)	(0.017)
High-value patents in the group	21,831	3,327	
Classes with high-value patents	64	25	

Notes: Estimator of Callaway and Sant'Anna (2021), doubly robust, with railway kilometers and the 1871 population × trend and the places whose chapter has not yet opened as the comparison group; the 493 places, 1877–1918. Outcomes: $\ln(1+x)$ of the high-value patents in the industrial classes (column 1) and in the traditional classes (column 2); column (3): $\ln(1+x_{\text{traditional}}) - \ln(1+x_{\text{industrial}})$ as its own outcome, so that its standard error is that of the difference of the two estimates. Industrial classes: all classes outside the traditional group. Panel A: the traditional classes of Table A2 and class 62 (salt works, 9 high-value patents). Panel B: the classes of Panel C and fertilizer (16), agriculture, forestry, horticulture, viticulture and dairy (45), and sugar and starch (89). Panel C: classes 2, 3, 6, 9, 11, 25, 28, 33, 34, 38, 41, 44, 51, 53, 54, 56, 66, 68, 69, 70, 71, 75, 77, 79, and 83, from bakery and clothing to toys and clocks (titles in Table A2). Years 6–12 and before founding: averages of event-study coefficients relative to the year before founding (years −11 to −2). High-value patents in the group: all high-value patents held in Germany, 1877–1918, with a class and a located holder (25,158); the estimates use the 493 places. Standard errors clustered by place in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

processes contribute the most, 8 percent of the additional high-value patents against 5 percent before the openings, followed by agriculture, mostly farm machinery, then steam engines and electrical engineering; no other class contributes more than about 5 percent.

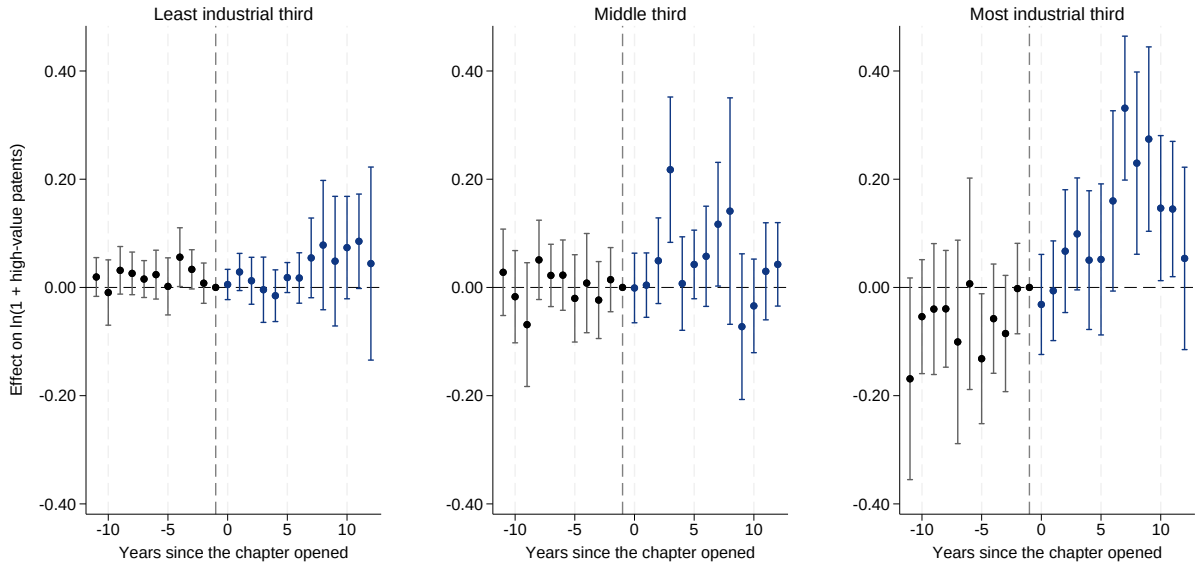
E.2. The Local Premium: All Designs, Selection on Renewal, and Classes

Among the high-value patents of the 22 territories, the rise of the local premium is 1.38 (0.46) without the railway and population controls, 1.45 to 2.13 when the 1, 3, 5 or 10 places with the most high-value patents are dropped and 1.06 to 1.79 when one territory at a time is dropped; in the industrial classes alone it is 1.32 (0.53) and in the traditional classes 0.54 (0.80) (Table E4 reports the further designs). For all places, the placebo estimates of the permutation test have a mean of 0.56, so part of the all-places estimate reflects a drift. With stack-by-class-by-year effects, the stacked estimate rises to 4.66 (2.26) on half the sample, only because the similarity to the same class elsewhere falls.

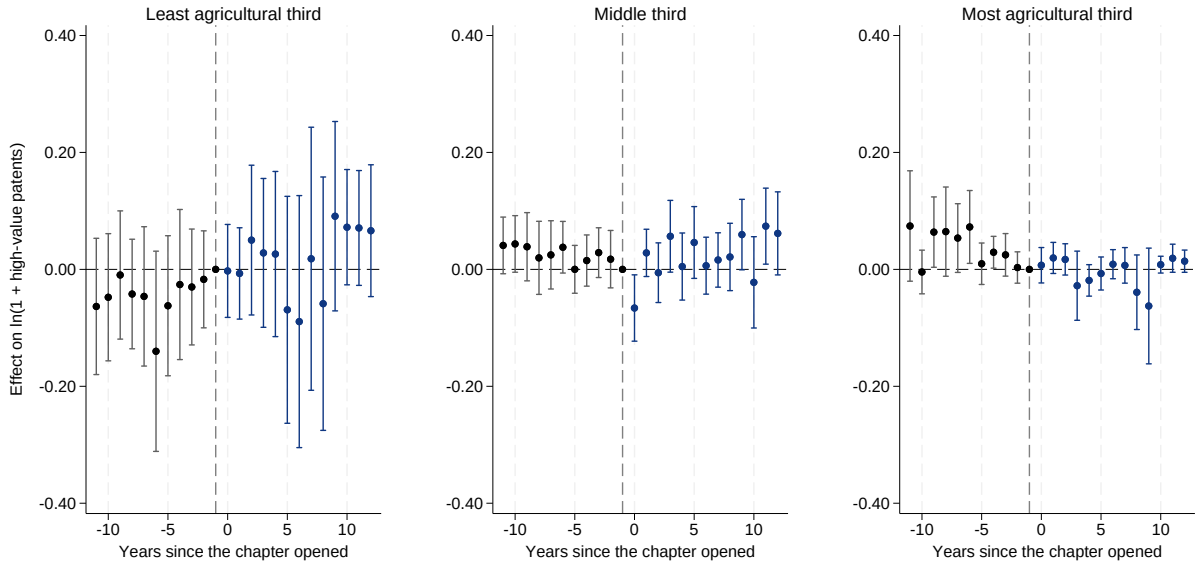
Selection on renewal accounts for about 40 percent of the estimate for high-value patents, and for about 70 percent at the upper end of its 95 percent confidence interval. Among all patents of the 22 territories with a local premium, the probability of renewal falls by 0.46 (0.34) percentage points per standard deviation of the local premium before an opening, and this relation rises by 1.33 (0.43) points after an opening ($p = 0.002$); without any change in the content of patents, a rise of 3.10 points per standard deviation would be needed to produce the estimate for high-value patents.

Figure E1. Event Studies by Third of the Industry Share on the Kreise and of the Agriculture Share

(a) Industry share, Kreise of 1882

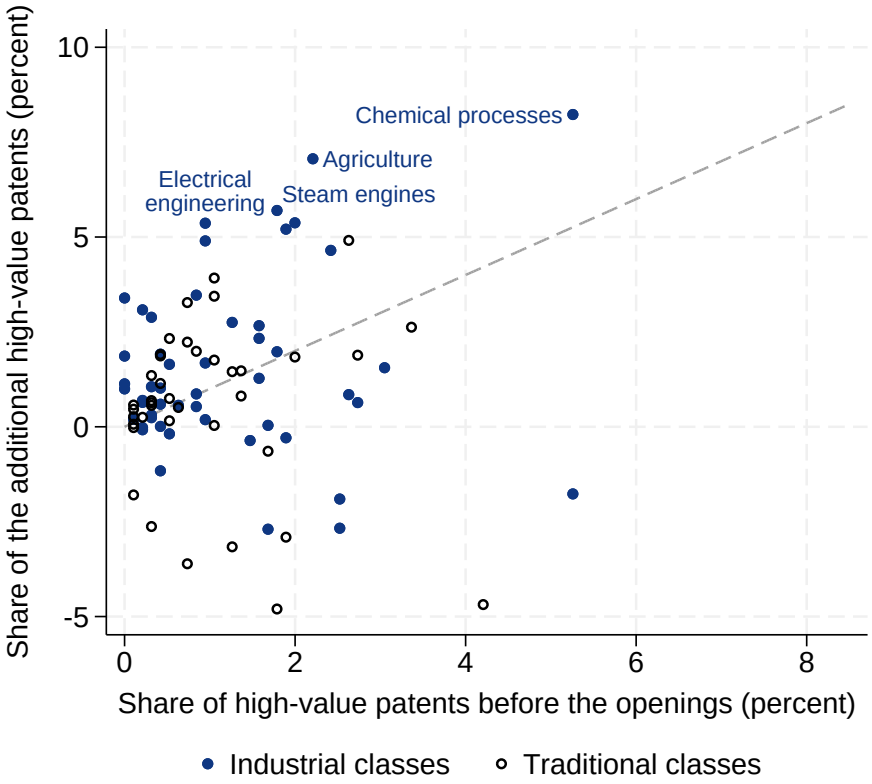


(b) Agriculture share, statistical units of 1900



Notes: Event-study coefficients on ln(1 + high-value patents), relative to the year before founding (dashed line; from the year of founding on in dark blue), with 95 percent confidence intervals, estimated separately within each third, each third against its own not-yet-treated places; estimator, covariates, and clustering as in Table E1.

Figure E2. Contribution of Each Patent Class to the Additional High-Value Patents



Notes: One dot per patent class: its share of the high-value patents in the not-yet-treated place-years (horizontal) against its share of the additional high-value patents after an opening (vertical). Class effects: estimator of Callaway and Sant'Anna (2021) on the count of high-value patents of the class, with railway kilometers and the 1871 population × trend; 493 places, 1877–1918; cells in which no place holds a patent of the class enter with an effect of zero. Classes on the dashed 45-degree line contributed in proportion to their size before the openings. Industrial and traditional classes as in Table A2. The class effects sum to 99 percent of the effect on all high-value patents, 0.171 (0.053) additional high-value patents per place and year.

Table E4. The Local Premium in the Embeddings under Alternative Designs, with Placebo Outcomes

	High-value patents		All register patents	
	22 chapters	All places	22 chapters	All places
Place and class × year effects (Table 4)	1.43^{***} (0.52)	1.17^{**} (0.52)	0.36 (0.28)	0.47^{*} (0.27)
permutation p-value	0.008	0.022		
firm families excluded	1.18^{***} (0.43)	1.06^{**} (0.44)		
industrial classes only	1.32^{**} (0.53)			
traditional classes only	0.54 (0.80)			
Stacked, clean controls, $K = 8$	1.27^{*} (0.77)	0.97 (0.73)	0.23 (0.44)	0.13 (0.42)
$K = 5$	0.68 (0.75)	0.27 (0.74)	0.08 (0.39)	−0.06 (0.38)
stack × class × year effects	4.66^{**} (2.26)	3.39^{*} (2.06)	0.16 (0.35)	0.06 (0.34)
Stacked event study: years −5 to −2		1.43 (0.95)		0.21 (0.37)
years 0 to 8		1.93^{**} (0.88)		0.26 (0.36)
Placebo: similarity to foreign patents	0.31 (0.38)	0.22 (0.32)	0.06 (0.17)	0.21 (0.15)
Placebo: similarity to other territories	0.31 (0.35)	0.11 (0.28)	0.08 (0.15)	0.13 (0.11)
Same territory, other places	−0.10 (0.45)	0.16 (0.37)	−0.16 (0.23)	0.12 (0.18)

Notes: Patent-level regressions as in Table 4. Local premium: mean cosine similarity of the embeddings to the patents of the place of the five preceding years minus that to the patents of the same class in other territories, ×100; same-holder pairs and addition patents excluded. Permutation: 500 random reassignments of founding years across territories. Firm families: pairs of holders that share a name token excluded. Industrial and traditional classes: the regression of the first row on the high-value patents of the 50 industrial (3,745 patents) and of the 38 traditional classes (1,565) of Table A2. Stacked: stacks as in Equation (B.1), with place and class × year effects within stacks or stack × class × year effects; the stacked samples include the never-treated places. Event study relative to the year before founding. Standard errors clustered by place in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix F. The Long Run and the Eighteenth-Century Economic Societies

The long-run comparisons of Section 8 ask a different question from the main design: not what an opening did to its territory, but whether the geography of invention that the chapters helped to shape persisted. Every West German county lay in a chapter territory by 1925, so the exercise is descriptive, in the manner of the persistence literature (Becker et al., 2016; Valencia Caicedo, 2019): it compares counties with the same population and density in 1871, the same distance to a large city and the same state, and sets the result against placebo rings around cities of the same size that had no chapter. It does not separate the chapters from the firms, engineering schools, and engineers that grew up with them and stayed in place.

All long-run designs use West German counties, because East and West Germany differed before the division (Becker et al., 2020b) and the East had a different patent system after 1945. The chapter rings hold 176 counties against 103, around 32 towns, and the placebo rings 151 against 83. The nearest-city comparison sets the counties whose nearest city of 20,000 inhabitants, 10 to 50 km away, was a chapter town by 1918 against those whose city was not, 89 against 81 counties, and also holds fixed the size of the city, early high-value patents and the 1882 industry. Standard errors are clustered by town or city.

The outcome, from PatentCity (Bergeaud and Verluise, 2024), is the number of patents granted to holders in the county in each of the periods 1877–1918, 1919–32, 1933–44, 1950–69, 1970–89, 1990–2009, and 2010–23, entered as $\ln(1 + \text{patents})$ as in the main design, with the county's population of 1871 among the controls; PatentCity's counts for 1877–1918 correlate with the high-value patents at 0.85 across counties. County GDP comes from the replication files of Peters (2022) and from the federal statistical office, and population from GPOP (Roesel, 2023). After 1980, PatentCity takes the locations of patents from de Rassenfosse et al. (2019).

F.1. Patents

Around chapter towns, a gap in patents appears between the wars and persists after 1950, while the placebo rings stay negative (Table F1); before 1918, no comparison in Panel A shows a clear gap. Between the wars the chapter rings show 0.28 (0.16); holding the 1882 industry fixed changes neither this gap nor the one after 1950, and the nearest-city comparison gives 0.21 (0.18) between the wars. Within counties, the change from before 1918 to after 1950 is 0.24 (0.10) in the chapter rings and 0.29 (0.15) in the nearest-city comparison, against −0.11 (0.15) in the placebo rings. In a county-period panel with county effects and state-period effects, the gap after 1950 exceeds that of the earlier periods by 0.13 (0.13), not significantly,

and it does not grow with the number of years for which a county's territory had a chapter by 1918. Around the towns whose chapter existed by 1877, the gap after 1950 is 0.51 (0.20), and around the West German towns of the chapters of the design 0.24 (0.14); the difference, −0.28 (0.21), is not significant ($p = 0.20$).

Density and distance to a large city reduce the early gap from 0.30 (0.11) to 0.09 (0.12) and the later gaps from 0.52 (0.15) and 0.60 (0.12) to 0.28 (0.16) and 0.32 (0.14) (Table F1, Panels A and B).

Table F1. Patents around Chapter Towns, Pooled by Era

	1877–1918	1919–44	1950–2023	After 1950 minus before 1918	Counties
Panel A. Population and density 1871, distance to a city of 50,000, state effects					
Chapter rings	0.087	0.284^{*}	0.323^{**}	0.236^{**}	279
	(0.120)	(0.160)	(0.135)	(0.101)	
+ 1882 industry and early patents	0.108	0.318^{**}	0.342^{**}		279
	(0.115)	(0.151)	(0.130)		
Nearest city, 1882 industry held fixed	0.070	0.204	0.358^{**}	0.287^{*}	170
	(0.105)	(0.182)	(0.149)	(0.147)	
Placebo rings	−0.091	−0.127	-0.200^{**}	−0.109	234
	(0.131)	(0.115)	(0.090)	(0.146)	
far from every chapter town	−0.133	−0.006	−0.201		82
	(0.132)	(0.154)	(0.134)		
Panel B. Population 1871 and state effects only					
Chapter rings	0.296^{**}	0.521^{***}	0.598^{***}		279
	(0.112)	(0.148)	(0.121)		
Nearest city	0.194	0.268	0.538^{***}		170
	(0.127)	(0.186)	(0.140)		

Notes: Outcome: ln(1 + patents granted in the period), averaged over the periods of the era (1919–44: two periods; 1950–2023: four); the county's population of 1871 is held fixed in every row. Chapter rings: counties 10–50 km against 50–100 km from a town whose chapter was founded by 1918, chapter-town counties dropped, standard errors clustered by town (32 towns). Placebo rings: the same around cities of 20,000 in 1871 without a chapter town within 12 km; far: only counties 50–100 km from every chapter town. Nearest city: counties whose nearest city of 20,000 lies 10–50 km away, chapter town against other city, with the city's 1871 size, the county's high-value patents of 1877–82, its 1882 industry share and its machine industry held fixed; standard errors clustered by city (59 cities). Fourth column: the within-county change, after 1950 minus before 1918, regressed on the same indicator and controls. Counties: the number in 1877–1918, 1990–2009, and 2010–23; in 1919–44, 271 to 272 (rings), 162 to 164 (nearest city), and 226 to 228 (placebo rings); in 1950–89, 274, 165, and 229. Patents: PatentCity (Bergeaud and Verluise, 2024). Stars from t distributions with $G - 1$ degrees of freedom, G the number of clusters: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F.2. Income and Population

The comparisons show no gap in income that survives the controls. Counties in the chapter rings had 12 to 14 log points more manufacturing GDP in 1935, 1961, and 1966, not signifi-

cantly, and with density and city distance held fixed every GDP coefficient lies between −2 and 2 log points; rings around cities without a chapter show similar gaps. The population of the chapter rings grew 27 (6) log points more from 1871 to 2019, and 11 (7) with density and city distance held fixed. Patents sit at the seat of their holder, so the gap in patents may reflect where inventive work was organized and owned.

F.3. The Economic Societies of the Eighteenth Century: Indicative Evidence

Cinnirella et al. (2025) show that the German regions with more members of the economic societies of the late eighteenth century were more innovative in the late nineteenth century, in least-squares estimates and with the distance to the nearest society seat as an instrument, and they trace this long-lasting effect to agglomeration economies and localized knowledge spillovers. Their Online Appendix documents the member data and an extensive set of robustness checks (Cinnirella et al., 2025, Online Appendix). With their data, their least-squares coefficient replicates exactly and their instrumental-variable estimate closely (first and third rows of Table F2).

Table F2 adds the Society's members of 1900, placed by coordinates, to their design. The exercise is only indicative. The two institutions differed in time, membership, and purpose: the economic societies were associations of the late Enlightenment that adopted, produced, and diffused useful knowledge, the Engineering Society a professional association whose chapters spread a century later. The members of both lived in many of the same places, so the data cannot separate their effects. And the Society's members of 1900 may in part be an outcome of the economic societies, through the earlier manufacturing, vocational schools, and skilled mechanics that Cinnirella et al. (2025) document, so holding them fixed may remove part of the effect of the economic societies rather than a confounder. With these caveats, the coefficient of the economic-society members becomes smaller when the Society's members of 1900 are added, in the least-squares and in the instrumental-variable estimates, and it remains positive and significant in both (Panel A). The Society's members of 1900 are themselves more numerous where the economic societies had more members (Panel B). The pattern is consistent with the long-lasting legacy that Cinnirella et al. (2025) document, one channel of which may have run through the engineers of 1900.

Table F2. Members of the Economic Societies, Society Members of 1900, and Patents in Grid Cells: Indicative Evidence

	Coefficient (SE) on		
Outcome, 2,698 grid cells of 15 km	Economic-society members	Society members of 1900	Chapter town within 30 km by 1900
Panel A. Data of Cinnirella et al. (2025)			
Patents 1877–1914	0.274^{***} (0.041)		
+ Society members of 1900	0.127^{***} (0.026)	0.572^{***} (0.023)	
IV, distance to the nearest society seat	0.314^{***} (0.109)		
+ Society members of 1900	0.210^{**} (0.093)	0.561^{***} (0.024)	
Panel B. High-value patents			
High-value patents 1877–1918	0.279^{***} (0.054)		
+ Society members of 1900	0.109^{***} (0.030)	0.661^{***} (0.032)	
Society members of 1900 as the outcome	0.257^{***} (0.047)		
High-value patents 1900–14, given 1877–99	0.067^{***} (0.023)		
+ Society members of 1900	0.032 (0.023)	0.314^{***} (0.034)	
+ chapter town within 30 km	0.036 (0.022)	0.298^{***} (0.036)	0.231^{***} (0.053)
IV, distance to the nearest society seat	0.250^{***} (0.094)		

Notes: Design of Cinnirella et al. (2025): grid cells of 15 km, their 15 controls, and 65 polity effects, standard errors clustered by polity, all variables in inverse hyperbolic sines. Economic-society members: members of the economic societies of the late eighteenth century, from their data (Cinnirella et al., 2025, Online Appendix). Society members of 1900: entries of the 1900 directory placed by coordinates (13,465). Their Table 2, column 3, reports 0.274 (0.040); their instrumental-variable estimate, column 6, is 0.329 (0.132). The estimates are indicative only (Appendix F.3). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.